

Jenkins Task

Create one Jenkins job using the below code and create six stages:

URL: <https://github.com/betawins/VProfile-1.git>

There are 6 stages in CI/CD :-

stage1: Git clone to download the source code.

 **Jenkins** / All ▾ / New Item

New Item

Enter an item name

Select an item type

 **Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

 **Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.

 **Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

 **Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

OK

 **Jenkins** / Task-01 ▾ / Configuration

Configure

 General
 **Source Code Management**
 Triggers
 Environment
 Build Steps
 Post-build Actions

☐ None

☒ **Git** ?

Repositories ?

Repository URL ?

Credentials ?

- none - ▾

+ Add

Advanced ▾

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

Save

Apply

Jenkins Task



Jenkins / Task-01 / #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Git Build Data

✓ #1 (Oct 27, 2025, 9:23:22 AM)



Started by user [admin](#)



This run spent:

- 26 ms waiting;
- 3.6 sec build duration;
- 3.6 sec total from scheduled to completion.



Revision: [afc137167b41aaafaeef82b81aee721ec883901](#)

Repository: <https://github.com/betawins/VProfile-1.git>

- refs/remotes/origin/master



No changes.

```
[root@ip-172-31-7-25 ~]# cd /var/lib/
lib/  local/ lock/  log/
[root@ip-172-31-7-25 ~]# cd /var/lib/jenkins/workspace/Task-01/
[root@ip-172-31-7-25 Task-01]# ls
Dockerfile  README.md  ansible  azure-pipelines.yml  pom.xml  src
[root@ip-172-31-7-25 Task-01]#
```

stage2: Sonarqube Integration to check the quality of code.

SonarQube Installation

SonarQube provides the capability to not only show health of an application but also to highlight issues newly introduced. With a Quality Gate in place, you can fix the leak and therefore improve code quality systematically.

#Prerequisites

1. EC2 instance with Java 1.8 installed [yum install java-1.8*]
2. Use t2.large with at least 20gb memory to run sonarqube.
3. MySQL Database Server or MySQL RDS instance.

Jenkins Task

EC2 > Instances > Launch an instance

Name and tags [Info](#)

Name

sonarqube

Add additional tags

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Q Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

▼ Storage (volumes) [Info](#)

Simple

EBS Volumes

Hide details

▼ Volume 1 (AMI Root) (Custom)

Storage type [Info](#)

EBS

Device name - required [Info](#)

/dev/xvda

Snapshot [Info](#)

snap-0374d739ea90e0ed5

Size (GiB) [Info](#)

20

Volume type [Info](#)

gp3

IOPS [Info](#)

3000

Delete on termination [Info](#)

Yes

Encrypted [Info](#)

Not encrypted

KMS key [Info](#)

Select

KMS keys are only applicable when encryption is set on this volume.

Throughput [Info](#)

125

Volume initialization rate - new, optional [Info](#)

Enter a value

Min: 100 MiB/s, Max: 300 MiB/s. [Additional charges apply](#)

```
root@ip-172-31-4-195:~
root@ip-172-31-4-195 ~]# java -version
penjdk version "1.8.0_462"
penJDK Runtime Environment Corretto-8.462.08.1 (build 1.8.0_462-b08)
penJDK 64-Bit Server VM Corretto-8.462.08.1 (build 25.462-b08, mixed mode)
root@ip-172-31-4-195 ~]#
```

Add mysql rpm Repository:-

- yum update
- sudo wget <https://dev.mysql.com/get/mysql57-community-release-el7-11.noarch.rpm>
- sudo yum localinstall mysql57-community-release-el7-11.noarch.rpm
- rpm --import <https://repo.mysql.com/RPM-GPG-KEY-mysql-2022>

Jenkins Task

- sudo yum install mysql-community-server
- sudo systemctl start mysqld.service
- sudo systemctl status mysqld.service

```
Complete!
[root@ip-172-31-4-195 ~]# sudo systemctl start mysqld.service
[root@ip-172-31-4-195 ~]# sudo systemctl status mysqld.service
● mysqld.service - MySQL Server
   Loaded: loaded (/usr/lib/systemd/system/mysqld.service; enabled; preset: disabled)
   Active: active (running) since Mon 2025-10-27 10:10:28 UTC; 8s ago
     Docs: man:mysqld(8)
           http://dev.mysql.com/doc/refman/en/using-systemd.html
   Process: 27339 ExecStartPre=/usr/bin/mysqld_pre_systemd (code=exited, status=0/SUCCESS)
   Process: 27391 ExecStart=/usr/sbin/mysqld --daemonize --pid-file=/var/run/mysqld/mysqld.pid $MYSQLD_OPTS (code=exited, status=0/SUCCESS)
   Main PID: 27393 (mysqld)
    Tasks: 27 (limit: 1053)
   Memory: 330.9M
      CPU: 2.865s
   CGroup: /system.slice/mysqld.service
           └─27393 /usr/sbin/mysqld --daemonize --pid-file=/var/run/mysqld/mysqld.pid

Oct 27 10:10:21 ip-172-31-4-195.ap-south-1.compute.internal systemd[1]: Starting mysqld.service - MySQL Server...
Oct 27 10:10:23 ip-172-31-4-195.ap-south-1.compute.internal mysqld_pre_systemd[27363]: mysqld: Out of memory (Needed 4294967200 bytes)
Oct 27 10:10:28 ip-172-31-4-195.ap-south-1.compute.internal systemd[1]: Started mysqld.service - MySQL Server.
lines 1-17/17 (END)
```

Configure the MySQL Root Password you will see default MySQL root password

grep 'temporary' /var/log/mysqld.log

!E2mwHsN=w?q Q&hPIED=y9kp

```
root@ip-172-31-4-195:~
[root@ip-172-31-4-195 ~]# grep 'temporary' /var/log/mysqld.log
2025-10-27T10:10:25.034864Z 1 [Note] A temporary password is generated for root@
localhost: !E2mwHsN=w?q
2025-10-27T10:10:27.961817Z 0 [Note] InnoDB: Creating shared tablespace for temporary tables
[root@ip-172-31-4-195 ~]# |
```

Login to mysql using the default password

mysql -u root -p

```
[root@ip-172-31-4-195 ~]# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 2
Server version: 5.7.44

Copyright (c) 2000, 2023, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> |
```

Now replace the default password with a new and strong password

- ALTER USER 'root'@'localhost' IDENTIFIED BY 'Admin@123';
- flush privileges;

Jenkins Task

```
mysql> ALTER USER 'root'@'localhost' IDENTIFIED BY 'Devops@0509';
Query OK, 0 rows affected (0.00 sec)

mysql> flush privileges;
Query OK, 0 rows affected (0.01 sec)

mysql> |
```

Exit from mysql and Test Using new password Devops@0509.

mysql -u root -p

```
mysql> exit
Bye
[root@ip-172-31-4-195 ~]# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 3
Server version: 5.7.44 MySQL Community Server (GPL)

Copyright (c) 2000, 2023, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> |
```

Download stable SonarQube version from below website.

Website: <https://www.sonarqube.org/downloads/>

Note: This Article written for SonarQube6.0

Download & unzip SonarQube 6.0

```
# cd /opt
```

```
# wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-6.6.zip
```

```
# unzip sonarqube-6.6.zip
```

```
# mv /opt/sonarqube-6.6 /opt/sonar
```

Jenkins Task

```
root@ip-172-31-4-195:~# cd /opt/
[root@ip-172-31-4-195 opt]# wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-6.6.zip
--2025-10-27 10:28:34-- https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-6.6.zip
Resolving binaries.sonarsource.com (binaries.sonarsource.com)... 108.159.61.118, 108.159.61.59, 108.159.61.119, ...
Connecting to binaries.sonarsource.com (binaries.sonarsource.com)|108.159.61.118|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 164350442 (157M) [application/zip]
Saving to: 'sonarqube-6.6.zip'

sonarqube-6.6.zip          100%[=====] 156.74M  31.5MB/s   in 5.8s

2025-10-27 10:28:40 (27.2 MB/s) - 'sonarqube-6.6.zip' saved [164350442/164350442]

[root@ip-172-31-4-195 opt]# ls
aws sonarqube-6.6.zip
[root@ip-172-31-4-195 opt]# unzip sonarqube-6.6.zip
Archive: sonarqube-6.6.zip
  creating: sonarqube-6.6/
    inflating: sonarqube-6.6/data/README.txt
    creating: sonarqube-6.6/logs/
    creating: sonarqube-6.6/elasticsearch/plugins/
[root@ip-172-31-4-195 opt]# ls
aws sonarqube-6.6 sonarqube-6.6.zip
[root@ip-172-31-4-195 opt]# mv /opt/sonarqube-6.6 /opt/sonar
[root@ip-172-31-4-195 opt]# ls
aws sonar sonarqube-6.6.zip
[root@ip-172-31-4-195 opt]# |
```

Login to mysql:-

```
mysql -u root -p
```

Create a new sonar database:-

```
CREATE DATABASE sonar CHARACTER SET utf8 COLLATE utf8_general_ci;
```

Create a local and a remote user:-

```
CREATE USER sonar@localhost IDENTIFIED BY 'Sonar@123';
```

```
CREATE USER sonar@'%' IDENTIFIED BY 'Sonar@123';
```

Grant database access permissions to users:-

```
GRANT ALL ON sonar.* TO sonar@localhost;
```

```
GRANT ALL ON sonar.* TO sonar@'%';
```

check users and databases:-

```
show databases;
```

```
SELECT User FROM mysql.user;
```

```
FLUSH PRIVILEGES;
```

```
QUIT
```

Jenkins Task

```
[root@ip-172-31-4-195 opt]# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 4
Server version: 5.7.44 MySQL Community Server (GPL)

Copyright (c) 2000, 2023, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE sonar CHARACTER SET utf8 COLLATE utf8_general_ci;
Query OK, 1 row affected (0.00 sec)

mysql> CREATE USER sonar@localhost IDENTIFIED BY 'Sonar@123';
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE USER sonar@%' IDENTIFIED BY 'Sonar@123';
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT ALL ON sonar.* TO sonar@localhost;
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT ALL ON sonar.* TO sonar@'%';
Query OK, 0 rows affected (0.00 sec)

mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sonar |
| sys |
+-----+
```

Jenkins Task

```
| sonar  
| sys  
+-----+  
5 rows in set (0.00 sec)  
  
mysql> SELECT User FROM mysql.user;  
+-----+  
| User  
+-----+  
| sonar  
| mysql.session  
| mysql.sys  
| root  
| sonar  
+-----+  
5 rows in set (0.01 sec)  
  
mysql> FLUSH PRIVILEGES;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> quit  
Bye  
[root@ip-172-31-4-195 opt]# |
```

So for you have configured required database information on mysql. Let's Jump back to your EC2 instance and enable SonarQube properties file to connect his Database.

ON EC2 Instance

Edit sonar properties file to uncomment and provide required information for below properties.

- File Name: /opt/sonar/conf/sonar.properties

```
root@ip-172-31-4-195:/opt/sonar/conf  
[root@ip-172-31-4-195 opt]# ls  
aws sonar sonarqube-6.6.zip  
[root@ip-172-31-4-195 opt]# cd sonar/  
[root@ip-172-31-4-195 sonar]# ls  
COPYING bin conf data elasticsearch extensions lib logs temp web  
[root@ip-172-31-4-195 sonar]# cd conf/  
[root@ip-172-31-4-195 conf]# ls  
sonar.properties wrapper.conf  
[root@ip-172-31-4-195 conf]# vi sonar.properties|
```

- sonar.jdbc.username=`sonar`

- sonar.jdbc.password=`Sonar@123`

Jenkins Task

-
sonar.jdbc.url=jdbc:mysql://localhost:3306/sonar?useUnicode=true&characterEncoding=utf8&rewriteBatchedStatements=true&useConfigs=maxPerformance&useSSL=false

- sonar.web.host='0.0.0.0'

- sonar.web.context='/sonar'

```
root@ip-172-31-4-195:/opt/sonar/conf
# Property values can:
# - reference an environment variable, for example sonar.jdbc.url= ${env:SONAR_JDBC_URL}
# - be encrypted. See https://redirect.sonarsource.com/doc/settings-encryption.html
#-----
# DATABASE
#
# IMPORTANT: the embedded H2 database is used by default. It is recommended for tests but not for
# production use. Supported databases are MySQL, Oracle, PostgreSQL and Microsoft SQLServer.
#
# User credentials.
# Permissions to create tables, indices and triggers must be granted to JDBC user.
# The schema must be created first.
sonar.jdbc.username=sonar
sonar.jdbc.password=sonar@123
#----- Embedded Database (default)
# H2 embedded database server listening port, defaults to 9092
#sonar.embeddedDatabase.port=9092
#----- MySQL 5.6 or greater
# Only InnoDB storage engine is supported (not myISAM).
# Only the bundled driver is supported. It can not be changed.
sonar.jdbc.url=jdbc:mysql://localhost:3306/sonar?useUnicode=true&characterEncoding=utf8&rewriteBatchedStatements=true&useConfigs=maxPerformance&useSSL=false
```

```
root@ip-172-31-4-195:/opt/sonar/conf
#
# Startup can be long if entropy source is short of entropy. Adding
# -Djava.security.egd=file:/dev/./urandom is an option to resolve the problem.
# See https://wiki.apache.org/tomcat/HowTo/FasterStartup#Entropy_Source
#
#sonar.web.javaOpts=-Xmx512m -Xms128m -XX:+HeapDumpOnOutOfMemoryError
#
# Same as previous property, but allows to not repeat all other settings like -Xmx
#sonar.web.javaAdditionalOpts=
#
# Binding IP address. For servers with more than one IP address, this property specifies which
# address will be used for listening on the specified ports.
# By default, ports will be used on all IP addresses associated with the server.
sonar.web.host=0.0.0.0
#
# Web context. When set, it must start with forward slash (for example /sonarqube).
# The default value is root context (empty value).
sonar.web.context=/sonar
#
# TCP port for incoming HTTP connections. Default value is 9000.
#sonar.web.port=9000
#
# The maximum number of connections that the server will accept and process at any given time.
# When this number has been reached, the server will not accept any more connections until
# the number of connections falls below this value. The operating system may still accept connections
# based on the sonar.web.connections.acceptCount property. The default value is 50.
#sonar.web.http.maxThreads=50
```

Switch to newly created user [ec2-user]

sudo su ec2-user

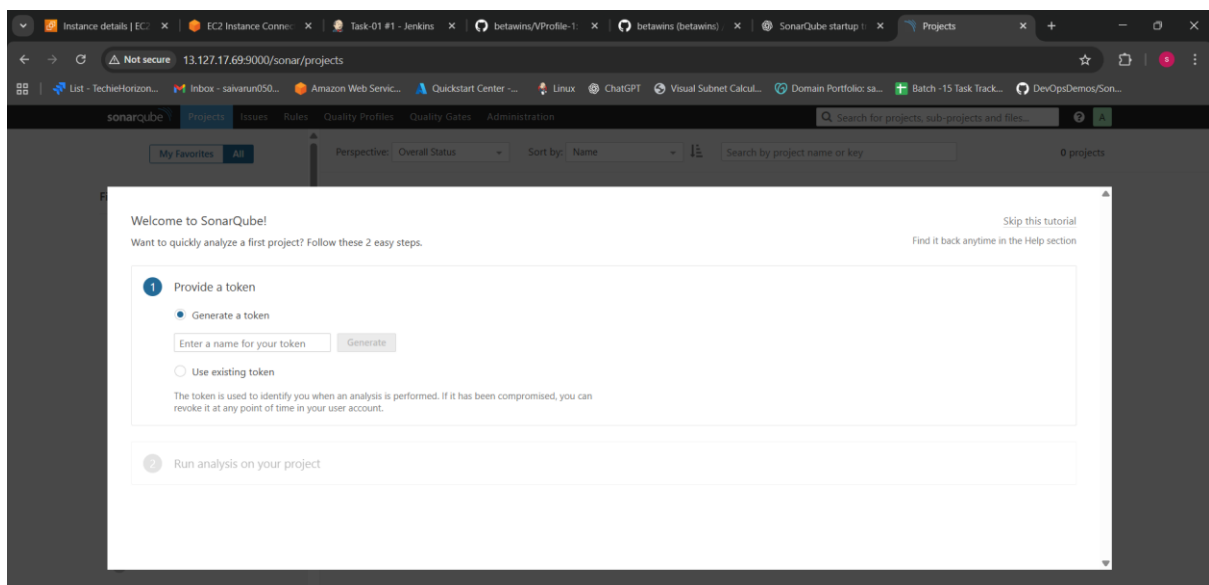
Start SonarQube service

cd /opt/sonar/bin/linux-x86-64/

./sonar.sh start

Jenkins Task

```
[root@ip-172-31-4-195 conf]# vi sonar.properties
[root@ip-172-31-4-195 conf]# exit
logout
[ec2-user@ip-172-31-4-195 ~]$ cd /opt/
[ec2-user@ip-172-31-4-195 opt]$ ll
total 160500
drwxr-xr-x.  4 root root          33 Oct 21 21:13 aws
drwxr-xr-x. 11 root root        141 Oct 20  2017 sonar
-rw-r--r--.  1 root root 164350442 Feb 16  2022 sonarqube-6.6.zip
[ec2-user@ip-172-31-4-195 opt]$ sudo chown -R ec2-user:ec2-user sonar
[ec2-user@ip-172-31-4-195 opt]$ ll
total 160500
drwxr-xr-x.  4 root      root          33 Oct 21 21:13 aws
drwxr-xr-x. 11 ec2-user ec2-user      141 Oct 20  2017 sonar
-rw-r--r--.  1 root      root 164350442 Feb 16  2022 sonarqube-6.6.zip
[ec2-user@ip-172-31-4-195 opt]$ cd sonar/bin/linux-x86-64/
[ec2-user@ip-172-31-4-195 linux-x86-64]$ ls
lib  sonar.sh  wrapper
[ec2-user@ip-172-31-4-195 linux-x86-64]$ ./sonar.sh start
Starting SonarQube...
started SonarQube.
[ec2-user@ip-172-31-4-195 linux-x86-64]$ ./sonar.sh status
SonarQube is not running.
[ec2-user@ip-172-31-4-195 linux-x86-64]$ ./sonar.sh status
SonarQube is not running.
[ec2-user@ip-172-31-4-195 linux-x86-64]$ ./sonar.sh start
Starting SonarQube...
started SonarQube.
[ec2-user@ip-172-31-4-195 linux-x86-64]$ ./sonar.sh status
```



<http://3.110.165.185:9000/sonar> - Path for sonarQube

User:-admin

password:-admin

Now add sonarqube scanner plugin in your Jenkins.

Jenkins Task

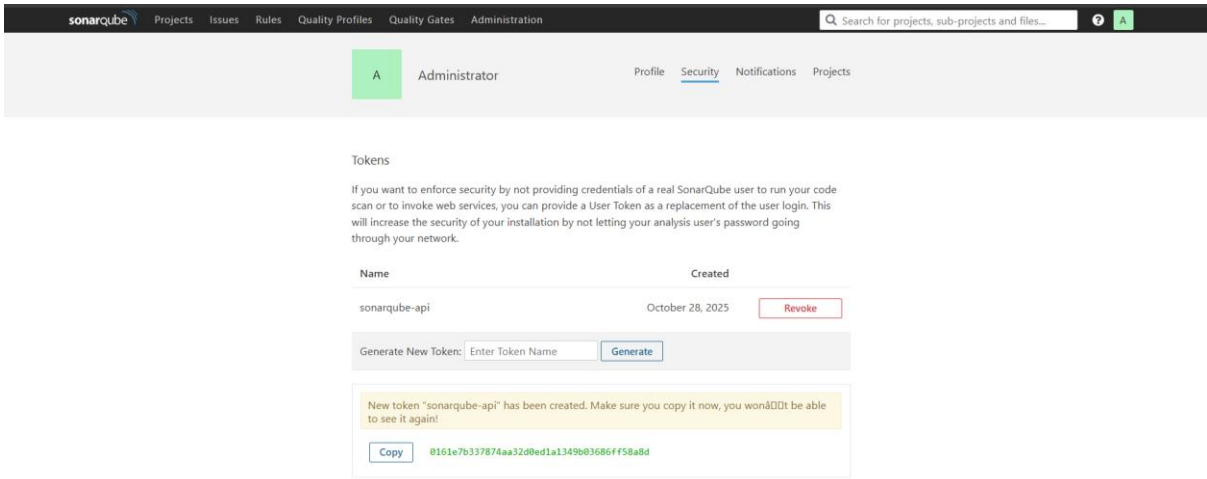


0161e7b337874aa32d0ed1a1349b03686ff58a8d

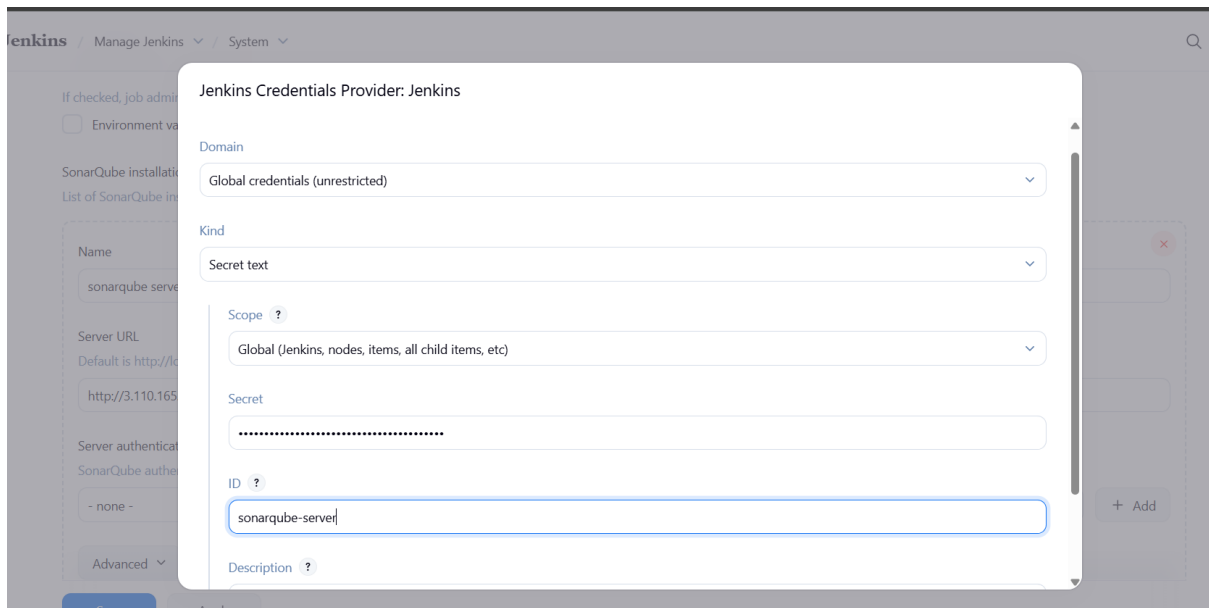
,

8c08b858482161f9e28d39a9b524d612ec8b8c04

Generate a key in sonarqube.



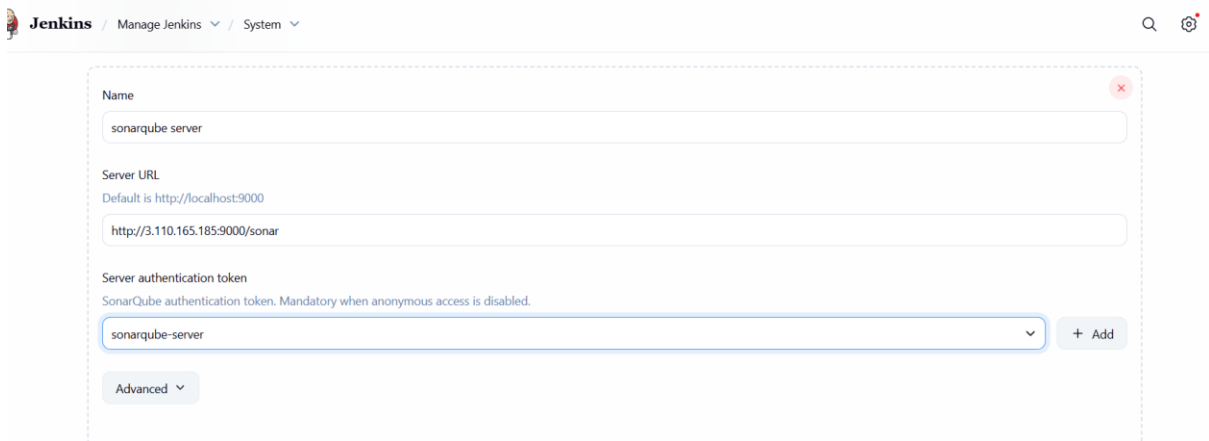
Jenkins Task



The screenshot shows the Jenkins 'Manage Jenkins' page with the 'System' tab selected. A modal window titled 'Jenkins Credentials Provider: Jenkins' is open. It contains the following fields:

- Domain:** A dropdown menu with 'Global credentials (unrestricted)' selected.
- Kind:** A dropdown menu with 'Secret text' selected.
- Scope:** A dropdown menu with 'Global (Jenkins, nodes, items, all child items, etc)' selected.
- Secret:** A text input field containing a series of dots, indicating a masked password.
- ID:** A text input field containing 'sonarqube-server'.
- Description:** A text input field, currently empty.

At the bottom right of the modal is a '+ Add' button. The background shows parts of the Jenkins configuration page, including 'SonarQube installation' and 'List of SonarQube instances'.



The screenshot shows the 'Jenkins' 'Manage Jenkins' page with the 'System' tab selected. A configuration box for a SonarQube instance is visible. It contains the following fields:

- Name:** A text input field containing 'sonarqube server'.
- Server URL:** A text input field containing 'http://3.110.165.185:9000/sonar'. Below the field, it says 'Default is http://localhost:9000'.
- Server authentication token:** A dropdown menu with 'sonarqube-server' selected. Below the field, it says 'SonarQube authentication token. Mandatory when anonymous access is disabled.'.
- Advanced:** A dropdown menu with a downward arrow.

At the bottom right of the configuration box is a '+ Add' button.

Now, sonar integration with Jenkins:-

Go to Jenkins and install sonar scanner

```
wget https://binaries.sonarsource.com/Distribution/sonar-scanner-cli/sonar-scanner-cli-4.6.2.2472-linux.zip
```

```
unzip sonar-scanner-cli-4.6.2.2472-linux.zip
```

```
mv sonar-scanner-4.6.2.2472-linux /opt/sonar_scanner
```

Jenkins Task

```
_/m/'
Last login: Tue Oct 28 07:26:51 2025 from 13.233.177.4
[ec2-user@ip-172-31-7-25 ~]$ sudo su -
Last login: Tue Oct 28 07:26:59 UTC 2025 on pts/1
[root@ip-172-31-7-25 ~]# systemctl start jenkins
[root@ip-172-31-7-25 ~]# cd /opt/
[root@ip-172-31-7-25 opt]# wget https://binaries.sonarsource.com/Distribution/sonar-scanner-cli/sonar-scanner-cli-4.6.2.2472-linux.zip
--2025-10-28 07:30:05-- https://binaries.sonarsource.com/Distribution/sonar-scanner-cli/sonar-scanner-cli-4.6.2.2472-linux.zip
Resolving binaries.sonarsource.com (binaries.sonarsource.com)... 108.159.61.119, 108.159.61.18, 108.159.61.59, ...
Connecting to binaries.sonarsource.com (binaries.sonarsource.com)|108.159.61.119|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 43099390 (41M) [application/zip]
Saving to: 'sonar-scanner-cli-4.6.2.2472-linux.zip'

sonar-scanner-cli-4 100%[=====>] 41.10M 51.5MB/s in 0.8s

2025-10-28 07:30:05 (51.5 MB/s) - 'sonar-scanner-cli-4.6.2.2472-linux.zip' saved
[43099390/43099390]

[root@ip-172-31-7-25 opt]# |
```

```
sonar-scanner-cli-4 100%[=====>] 41.10M 51.5MB/s in 0.8s

2025-10-28 07:30:05 (51.5 MB/s) - 'sonar-scanner-cli-4.6.2.2472-linux.zip' saved
[43099390/43099390]

[root@ip-172-31-7-25 opt]# ls
aws sonar-scanner-cli-4.6.2.2472-linux.zip
[root@ip-172-31-7-25 opt]# unzip sonar-scanner-cli-4.6.2.2472-linux.zip
Archive: sonar-scanner-cli-4.6.2.2472-linux.zip
  creating: sonar-scanner-4.6.2.2472-linux/
  creating: sonar-scanner-4.6.2.2472-linux/jre/
```

```
sonar-scanner-4.6.2.2472-linux/jre/legal/java.management.rmi/ASSEMBLY_EXCEPTION -> ../java.base/ASSEMBLY_EXCEPTION
sonar-scanner-4.6.2.2472-linux/jre/legal/jdk.jsobject/ADDITIONAL_LICENSE_INFO -> ../java.base/ADDITIONAL_LICENSE_INFO
sonar-scanner-4.6.2.2472-linux/jre/legal/jdk.jsobject/LICENSE -> ../java.base/LICENSE
sonar-scanner-4.6.2.2472-linux/jre/legal/jdk.jsobject/ASSEMBLY_EXCEPTION -> ../java.base/ASSEMBLY_EXCEPTION
[root@ip-172-31-7-25 opt]# ls
aws sonar-scanner-4.6.2.2472-linux sonar-scanner-cli-4.6.2.2472-linux.zip
[root@ip-172-31-7-25 opt]# mv sonar-scanner-4.6.2.2472-linux /opt/sonar_scanner
[root@ip-172-31-7-25 opt]# ls
aws sonar-scanner-cli-4.6.2.2472-linux.zip sonar_scanner
[root@ip-172-31-7-25 opt]# |
```

Set SonarQube server details in sonar-scanner property file

Sonar properties file: /opt/sonar_scanner/conf/sonar-scanner.properties

sonar.host.url=http://`<SONAR\_SERVER\_IP>`:9000

```
[root@ip-172-31-7-25 opt]# ls
aws sonar-scanner-cli-4.6.2.2472-linux.zip sonar_scanner
[root@ip-172-31-7-25 opt]# cd sonar_scanner/
[root@ip-172-31-7-25 sonar_scanner]# ls
bin conf jre lib
[root@ip-172-31-7-25 sonar_scanner]# cd conf/
[root@ip-172-31-7-25 conf]# ls
sonar-scanner.properties
[root@ip-172-31-7-25 conf]# vi sonar-scanner.properties|
```

Jenkins Task

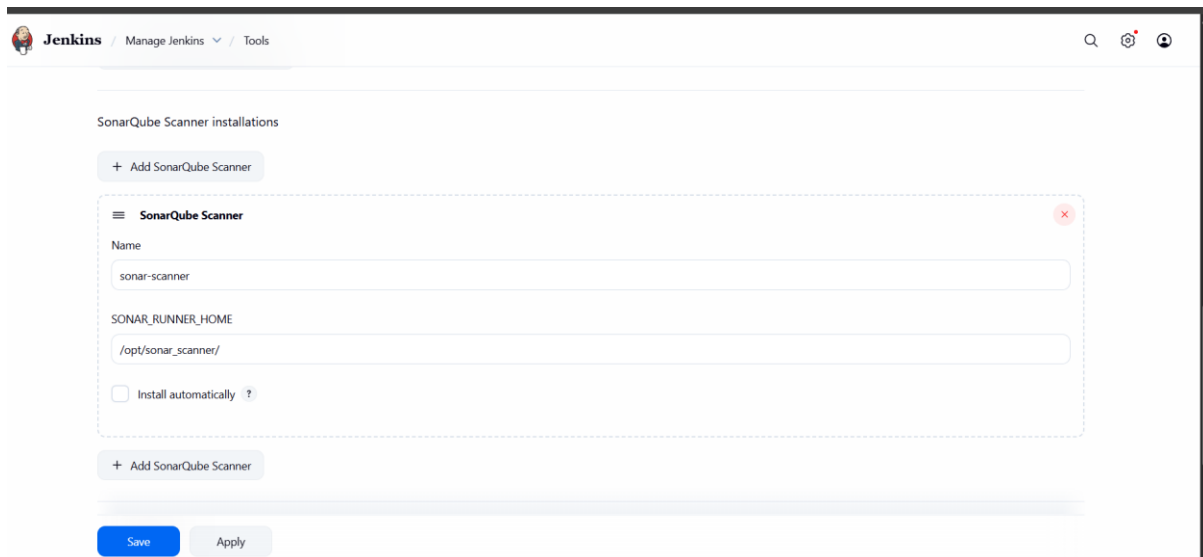
```
root@ip-172-31-7-25:/opt/sonar_scanner/conf
#Configure here general information about the environment, such as SonarQube server connection details for example
#No information about specific project should appear here

#----- Default SonarQube server
sonar.host.url=http://3.110.165.185:9000/sonar

#----- Default source code encoding
#sonar.sourceEncoding=UTF-8

~
~
~
```

```
[root@ip-172-31-7-25 conf]# cd /opt/sonar_scanner/
[root@ip-172-31-7-25 sonar_scanner]# ls
bin  conf  jre  lib
[root@ip-172-31-7-25 sonar_scanner]# cd bin/
[root@ip-172-31-7-25 bin]# ls
sonar-scanner  sonar-scanner-debug
[root@ip-172-31-7-25 bin]# pwd
/opt/sonar_scanner/bin
[root@ip-172-31-7-25 bin]# |
```



The screenshot shows the Jenkins web interface for configuring a SonarQube Scanner. The page title is "SonarQube Scanner installations". There is a button "+ Add SonarQube Scanner" at the top. Below it, a configuration card for "SonarQube Scanner" is shown. The card has a red close button in the top right corner. Inside the card, there are two input fields: "Name" with the value "sonar-scanner" and "SONAR_RUNNER_HOME" with the value "/opt/sonar_scanner/". Below these fields is a checkbox labeled "Install automatically" which is currently unchecked. At the bottom of the card is another "+ Add SonarQube Scanner" button. Below the configuration card, there are two buttons: "Save" and "Apply".

sonar.projectKey=Sabear

sonar.projectName=Sabear

sonar.projectVersion=1.0

sonar.sources=/var/lib/jenkins/workspace/\$JOB_NAME/src/

sonar.binaries=target/classes/com/visualpathit/account/controller/

sonar.junit.reportsPath=target/surefire-reports

sonar.jacoco.reportPath=target/jacoco.exec

sonar.java.binaries=src/main/java/com/visualpathit/account/

Jenkins Task

Jenkins

Task-01

Configuration

Search

Help

User

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute SonarQube Scanner

JDK ?
JDK to be used for this SonarQube analysis
(Inherit From Job)

Path to project properties ?

Analysis properties ?
sonar.projectKey=Sabear
sonar.projectName=Sabear
sonar.projectVersion=1.0
sonar.sources=/var/lib/jenkins/workspace/SJOB_NAME/src/
sonar.binaries=target/classes/com/visualpathit/account/controller/
sonar.junit.reportsPath=target/surefire-reports

Additional arguments ?

Save Apply

Jenkins

Task-01

Status

Changes

Workspace

Build Now

Configure

Delete Project

SonarQube

Rename

Credentials

Task-01

v-profile

SonarQube

Permalinks

Last build (#2), 1 hr 30 min ago

Last stable build (#2), 1 hr 30 min ago

Last successful build (#2), 1 hr 30 min ago

Last completed build (#2), 1 hr 30 min ago

Builds

Filter

Today

Jenkins Task



Jenkins / Task-01 ▾ / Changes



Workspace



Build Now



Configure



Delete Project



SonarQube



Rename



Credentials

Builds



Filter



Today



#3 7:53 AM



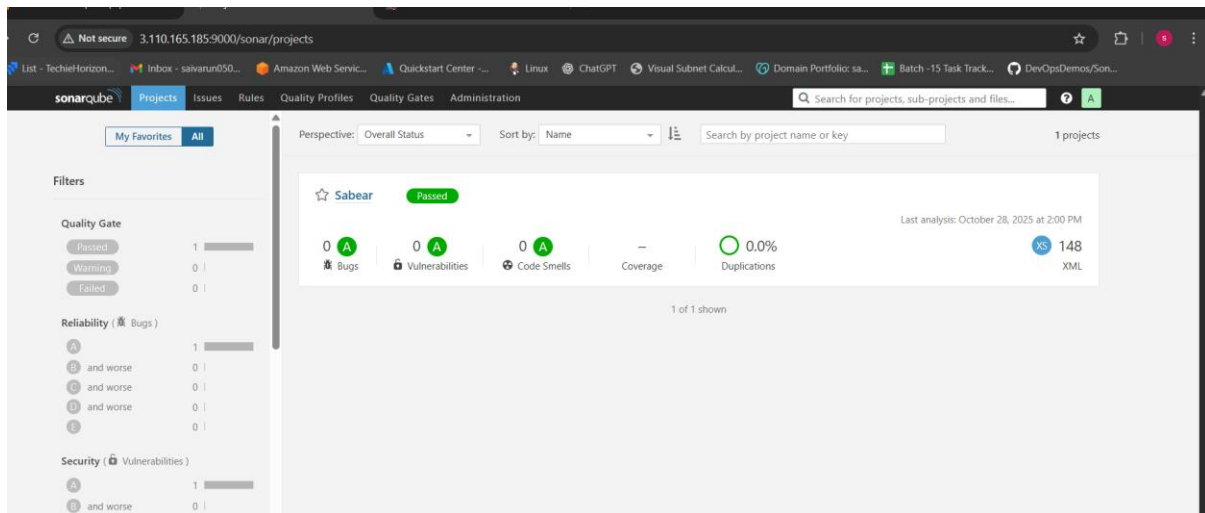
#2 6:22 AM



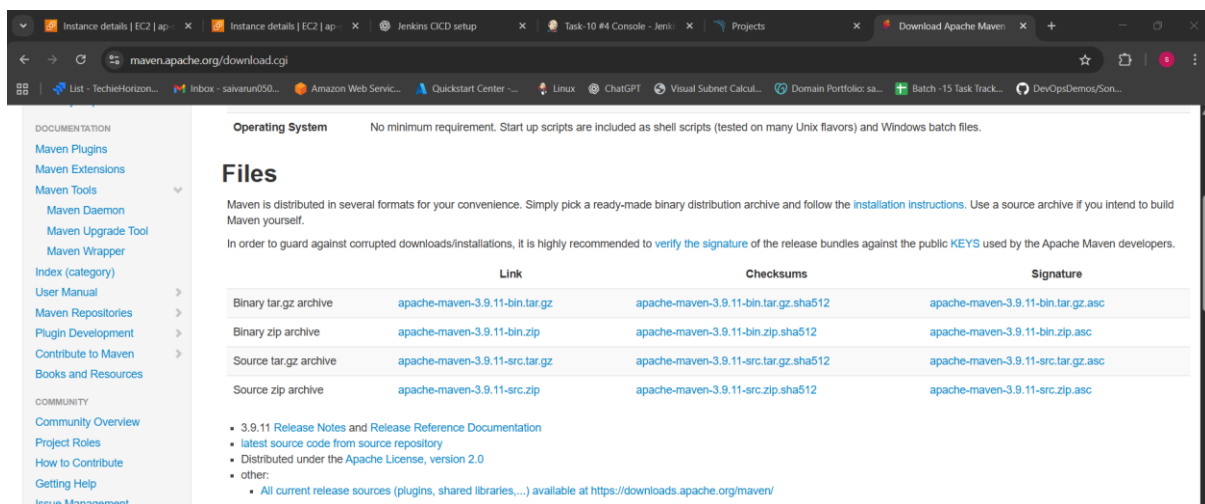
October 27, 2025

```
INFO: Sensor CPD Block Indexer (done) | time=67ms
INFO: SCM provider for this project is: git
INFO: 23 files to be analyzed
INFO: 23/23 files analyzed
INFO: 6 files had no CPD blocks
INFO: Calculating CPD for 10 files
INFO: CPD calculation finished
INFO: Analysis report generated in 166ms, dir size=109 KB
INFO: Analysis reports compressed in 38ms, zip size=48 KB
INFO: Analysis report uploaded in 238ms
INFO: ANALYSIS SUCCESSFUL, you can browse http://3.110.165.185:9000/sonar/dashboard/index/Sabear
INFO: Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
INFO: More about the report processing at http://3.110.165.185:9000/sonar/api/ce/task?id=AZop8Fkf5pT7rGoW\_-yr
INFO: Task total time: 7.819 s
INFO: -----
INFO: EXECUTION SUCCESS
INFO: -----
INFO: Total time: 10.174s
INFO: Final Memory: 10M/118M
INFO: -----
Finished: SUCCESS
```

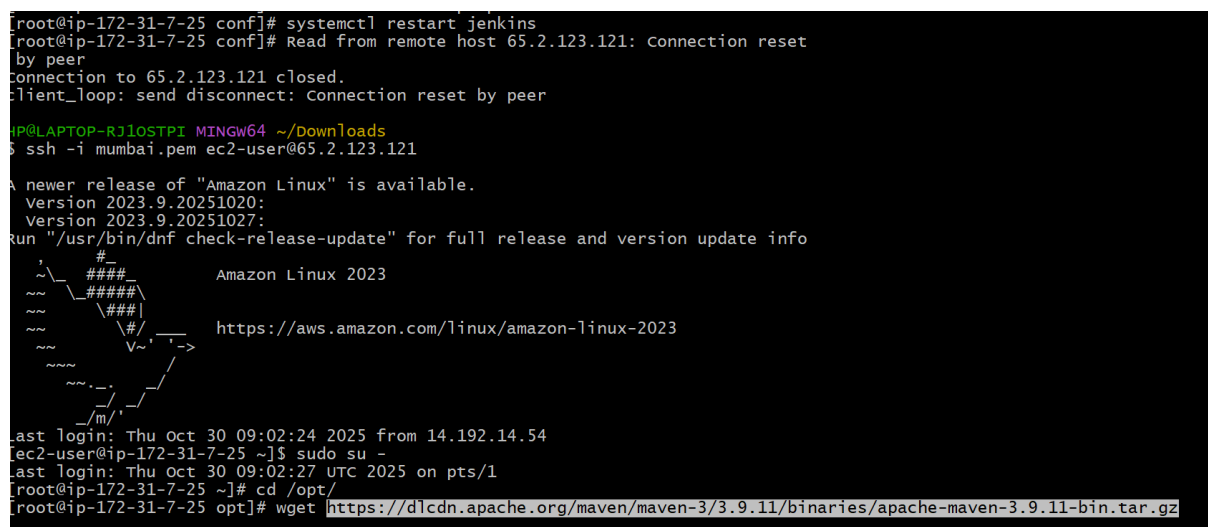

Jenkins Task



Go to Jenkins instance and download **maven** in it.



<https://dlcdn.apache.org/maven/maven-3/3.9.11/binaries/apache-maven-3.9.11-bin.tar.gz>



Jenkins Task

```
[root@ip-172-31-7-25 opt]# wget https://dlcdn.apache.org/maven/maven-3/3.9.11/binaries/apache-maven-3.9.11-bin.tar.gz
--2025-10-30 09:47:02-- https://dlcdn.apache.org/maven/maven-3/3.9.11/binaries/apache-maven-3.9.11-bin.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9160848 (8.7M) [application/x-gzip]
Saving to: 'apache-maven-3.9.11-bin.tar.gz'

apache-maven-3.9.11-bin.tar.gz  100%[=====>] 8.74M  --.-KB/s  in 0.05s

2025-10-30 09:47:02 (163 MB/s) - 'apache-maven-3.9.11-bin.tar.gz' saved [9160848/9160848]

[root@ip-172-31-7-25 opt]# ls
apache-maven-3.9.11-bin.tar.gz  aws  sonar-scanner-cli-4.6.2.2472-linux.zip  sonar_scanner
[root@ip-172-31-7-25 opt]# |
```

tar xvf apache-maven-3.9.11-bin.tar.gz

Extract the tar file by tar xvf

```
apache-maven-3.9.11/lib/commons-cli-1.9.0.jar
apache-maven-3.9.11/lib/maven-model-builder-3.9.11.jar
[root@ip-172-31-7-25 opt]# ls
apache-maven-3.9.11  apache-maven-3.9.11-bin.tar.gz  aws  sonar-scanner-cli-4.6.2.2472-linux.zip  sonar_scanner
[root@ip-172-31-7-25 opt]# |
```

```
apache-maven-3.9.11/ aws/
[root@ip-172-31-7-25 opt]# cd apache-maven-3.9.11/
[root@ip-172-31-7-25 apache-maven-3.9.11]# ls
LICENSE  NOTICE  README.txt  bin  boot  conf  lib
[root@ip-172-31-7-25 apache-maven-3.9.11]# cd bin/
[root@ip-172-31-7-25 bin]# ls
m2.conf  mvn  mvn.cmd  mvnDebug  mvnDebug.cmd  mvnyjp
[root@ip-172-31-7-25 bin]# pwd
/opt/apache-maven-3.9.11/bin
[root@ip-172-31-7-25 bin]# |
```

```
apache-maven-3.9.11/lib/wagon-http-shared-3.5.3.jar
apache-maven-3.9.11/lib/httpclient-4.5.14.jar
apache-maven-3.9.11/lib/maven-core-3.9.11.jar
apache-maven-3.9.11/lib/commons-cli-1.9.0.jar
[root@ip-172-31-7-25 ~]# vi .bash_profile
[root@ip-172-31-7-25 ~]# |
```

```
root@ip-172-31-7-25:~
# .bash_profile

# Get the aliases and functions
if [ -f ~/.bashrc ]; then
    . ~/.bashrc
fi

# User specific environment and startup programs
MVN_HOME=/opt/apache-maven-3.9.11/bin
PATH=$PATH:$HOME/bin:$MVN_HOME

export PATH
```

Jenkins Task

```
apache-maven-3.9.11/bin/commons-cli-1.9.0.jar
[root@ip-172-31-7-25 ~]# vi .bash_profile
[root@ip-172-31-7-25 ~]# source ~/.bash_profile
[root@ip-172-31-7-25 ~]# echo $MVN_HOME
/opt/apache-maven-3.9.11/bin
[root@ip-172-31-7-25 ~]#
```

```
[root@ip-172-31-7-25 ~]# mvn --version
Apache Maven 3.9.11 (3e54c93a704957b63ee3494413a2b544fd3d825b)
Maven home: /opt/apache-maven-3.9.11
Java version: 17.0.16, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-17-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.1.155-176.282.amzn2023.x86_64", arch: "amd64", family: "unix"
[root@ip-172-31-7-25 ~]#
```

The screenshot shows the Jenkins 'Maven installations' configuration page. At the top, there's a header with the Jenkins logo, 'Manage Jenkins', and 'Tools'. Below this, a 'Maven installations' section contains a '+ Add Maven' button and a configuration card for a Maven installation named 'Maven'. The card has a red 'X' icon in the top right corner. Inside the card, there are input fields for 'Name' (containing 'MVN_HOME') and 'MAVEN_HOME' (containing '/opt/apache-maven-3.9.11'). There is also a checkbox for 'Install automatically' which is currently unchecked. Below the configuration card, there is another '+ Add Maven' button. At the bottom of the page, there are 'Save' and 'Apply' buttons.

The screenshot shows the Jenkins 'Task-03' configuration page. The header includes the Jenkins logo, 'Task-03', and 'Configuration'. On the left side, there is a sidebar with a 'Configure' section and a list of tabs: 'General', 'Source Code Management', 'Triggers', 'Environment', 'Build Steps', and 'Post-build Actions'. The 'Build Steps' tab is currently selected. The main content area shows the configuration for the 'Build Steps' tab. It includes a section for 'Analysis properties' with a text area containing the following properties:
sonar.projectKey=Sabear
sonar.projectName=Sabear
sonar.projectVersion=1.0
sonar.sources=/var/lib/jenkins/workspace/\$JOB_NAME/src/
sonar.binaries=target/classes/com/visualpathit/account/controller/
sonar.junit.reportsPath=target/surefire-reports
sonar.jacoco.reportPath=target/jacoco.exec
sonar.java.binaries=src/main/java/com/visualpathit/account/
Below this, there are sections for 'Additional arguments' and 'JVM Options', each with a text input field and a dropdown arrow. At the bottom, there is a '+ Add build step' button.

Jenkins Task

Add built step -> Invoke top-level Maven targets

ns / Task-03 / Configuration

JVM Options

Code Management

Environment

Actions

Invoke top-level Maven targets

Maven Version

Goals

clean install

Advanced

+ Add build step

```
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.jar (193 kB at 4.1 MB/s)
[INFO] Installing /var/lib/jenkins/workspace/Task-03/pom.xml to /var/lib/jenkins/.m2/repository/com/visualpathit/vprofile/v2/vprofile-v2.pom
[INFO] Installing /var/lib/jenkins/workspace/Task-03/target/vprofile-v2.war to /var/lib/jenkins/.m2/repository/com/visualpathit/vprofile/v2/vprofile-v2.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 8.546 s
[INFO] Finished at: 2025-10-30T11:07:23Z
[INFO] -----
Finished: SUCCESS
```

```
[root@ip-172-31-7-25 ~]# systemctl restart jenkins
[root@ip-172-31-7-25 ~]# cd /var/lib/jenkins/workspace/
[root@ip-172-31-7-25 workspace]# ls
Task-01 Task-02 Task-03 Task-10 first_job
[root@ip-172-31-7-25 workspace]# cd Task-03/
[root@ip-172-31-7-25 Task-03]# ls
Dockerfile README.md ansible azure-pipelines.yml pom.xml src target
[root@ip-172-31-7-25 Task-03]# |
```

Target file is created as above. Till here maven process is done.

NEXUS:-

Create one ec2 for nexus.

Jenkins Task

EC2 > Instances > i-Oda7b84f6449b5c6d

Instance summary for i-Oda7b84f6449b5c6d (Nexus) info

Updated 7 minutes ago

Instance ID
i-Oda7b84f6449b5c6d

IPv6 address
-

Hostname type
IP name: ip-172-31-28-159.ap-south-1.compute.internal

Answer private resource DNS name
IPv4 (A)
65.0.116.164 [Public IP]

Auto-assigned IP address
65.0.116.164 [Public IP]

IAM Role
-

IMDSv2
Required

Public IPv4 address
65.0.116.164 | [open address](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-28-159.ap-south-1.compute.internal

Instance type
c7i-flex.large

VPC ID
vpc-03b94da1c23d84183

Subnet ID
subnet-0cdf12a5d9e8086ff

Instance ARN
arn:aws:ec2:ap-south-1:868011652978:instance/i-Oda7b84f6449b5c6d

Private IPv4 addresses
172.31.28.159

Public DNS
ec2-65-0-116-164.ap-south-1.compute.amazonaws.com | [open address](#)

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name
-

Managed
false

yum -y install java-1.8*

```
[ec2-user@ip-172-31-28-159 ~]$ sudo su -
[root@ip-172-31-28-159 ~]# yum -y install java-1.8*
Amazon Linux 2023 Kernel Livepatch repository 300 kB/s | 28 kB 00:00
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
java-1.8.0-amazon-corretto x86_64 1:1.8.0_472.b08-1.amzn2023 amazonlinux 38 M
java-1.8.0-amazon-corretto-devel x86_64 1:1.8.0_472.b08-1.amzn2023 amazonlinux 63 M
Installing dependencies:
adwaita-cursor-theme noarch 47.0-1.amzn2023.0.1 amazonlinux 325 k
adwaita-icon-theme noarch 47.0-1.amzn2023.0.1 amazonlinux 285 k
```

cd /opt

sudo wget https://download.sonatype.com/nexus/3/nexus-3.68.1-02-unix.tar.gz

sudo tar -xvf nexus-3.68.1-02-unix.tar.gz

mv /opt/nexus-3.68.1-02 /opt/nexus

```
[root@ip-172-31-28-159 opt]# sudo wget https://download.sonatype.com/nexus/3/nexus-3.68.1-02-unix.tar.gz
--2025-10-30 11:49:29-- https://download.sonatype.com/nexus/3/nexus-3.68.1-02-unix.tar.gz
Resolving download.sonatype.com (download.sonatype.com)... 52.53.133.15, 52.52.158.75
Connecting to download.sonatype.com (download.sonatype.com)|52.53.133.15|:443... connected.
HTTP request sent, awaiting response... 302 Moved Temporarily
Location: https://cdn.download.sonatype.com/repository/downloads-prod-group/3/nexus-3.68.1-02-unix.tar.gz [following]
```

Jenkins Task

```
nexus-3.68.1-02-unix 100%[=====>] 218.13M 158MB/s in 1.4s
2025-10-30 11:49:32 (158 MB/s) - 'nexus-3.68.1-02-unix.tar.gz' saved [228724143/228724143]

[root@ip-172-31-28-159 opt]# sudo tar -xvf nexus-3.68.1-02-unix.tar.gz
nexus-3.68.1-02/.install4j/9d17dc87.lprop
nexus-3.68.1-02/.install4j/MessagesDefault
nexus-3.68.1-02/.install4j/build.uuid
nexus-3.68.1-02/.install4j/e4ada6b7.lprop
nexus-3.68.1-02/.install4j/i4j_extf_0_17islik.utf8
nexus-3.68.1-02/.install4j/i4j_extf_10_17islik_10358jn.png
nexus-3.68.1-02/.install4j/i4j_extf_11_17islik_1qne9sv.png
```

```
sonatype-work/nexus3/log/.placeholder
sonatype-work/nexus3/tmp/.placeholder
[root@ip-172-31-28-159 opt]# mv /opt/nexus-3.68.1-02 /opt/nexus
[root@ip-172-31-28-159 opt]# ls
aws nexus nexus-3.68.1-02-unix.tar.gz sonatype-work
[root@ip-172-31-28-159 opt]# |
```

```
[root@ip-172-31-28-159 ~]# cd /opt/
[root@ip-172-31-28-159 opt]# ls
aws nexus nexus-3.68.1-02-unix.tar.gz sonatype-work
[root@ip-172-31-28-159 opt]# cd nexus/
[root@ip-172-31-28-159 nexus]# ls
NOTICE.txt      PRO-LICENSE.txt  deploy  lib      replicator
OSS-LICENSE.txt bin              etc     public   system
[root@ip-172-31-28-159 nexus]# cd bin/
[root@ip-172-31-28-159 bin]# vi nexus.
nexus.rc        nexus.vmoptions
[root@ip-172-31-28-159 bin]# vi nexus.rc
```

```
root@ip-172-31-28-159:/opt/nexus/bin
run_as_user="ec2-user"
~
~
~
~
~
~
~
~
~
~
```

Save and exit from file.

Then exit from root user and switch to ec2-user.

Jenkins Task

And change the permission.

Cd ~

Cd /opt/

Sudo chown -R ec2-user:ec2-user nexus

```
nexus is stopped.  
ec2-user@ip-172-31-28-159 bin]$ cd /opt/  
ec2-user@ip-172-31-28-159 opt]$ sudo chown -R ec2-user:ec2-user nexus
```

```
[ec2-user@ip-172-31-28-159 bin]$ cd /opt/  
[ec2-user@ip-172-31-28-159 opt]$ ll  
total 223364  
drwxr-xr-x.  4 root      root          33 Oct 23 00:08 aws  
drwxr-xr-x. 10 ec2-user  ec2-user      181 Oct 30 11:49 nexus  
-rw-r--r--.  1 root      root      228724143 May 16  2024 nexus-3.68.1-02-unix.tar.gz  
drwxr-xr-x.  3 root      root          20 Oct 30 11:49 sonatype-work  
[ec2-user@ip-172-31-28-159 opt]$ |
```

sudo mkdir -p /opt/sonatype-work/nexus3/log

sudo chown -R ec2-user:ec2-user /opt/nexus /opt/sonatype-work

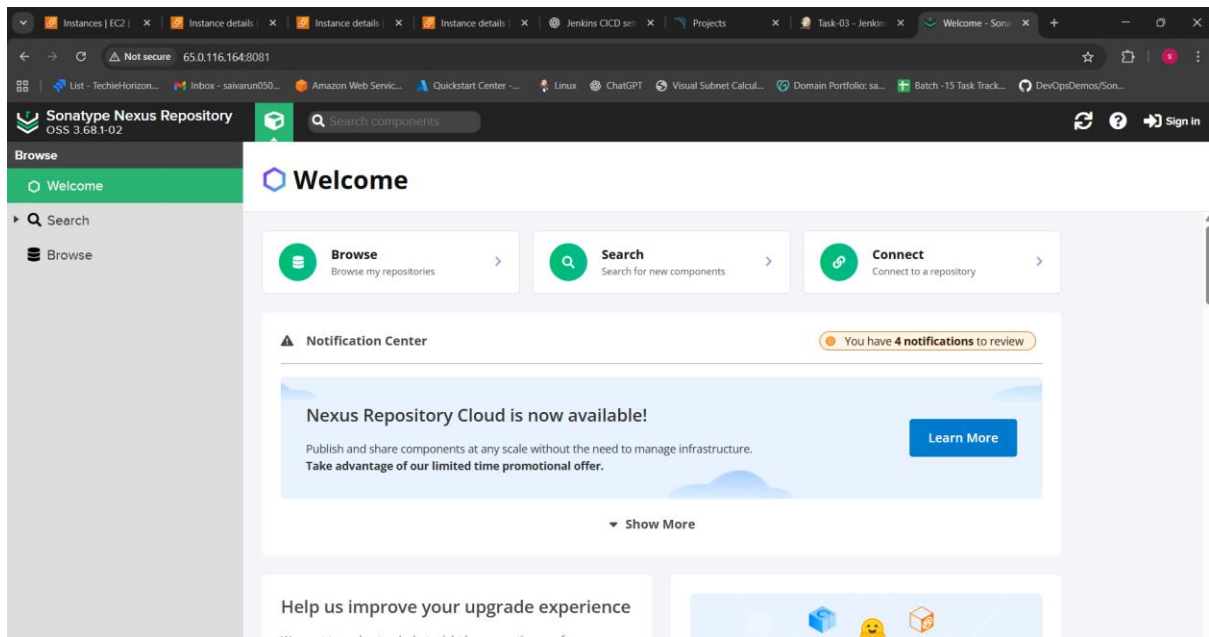
sudo chmod -R 755 /opt/nexus /opt/sonatype-work

If you are facing like nexus started but the status is showing not running. Then use above 3 commands in ec2-user .

```
[ec2-user@ip-172-31-28-159 ~]$ cd /opt/nexus/bin/  
[ec2-user@ip-172-31-28-159 bin]$ ./nexus start  
Starting nexus  
[ec2-user@ip-172-31-28-159 bin]$ ./nexus status  
nexus is stopped.  
[ec2-user@ip-172-31-28-159 bin]$ cd ~  
[ec2-user@ip-172-31-28-159 ~]$ sudo mkdir -p /opt/sonatype-work/nexus3/log  
sudo chown -R ec2-user:ec2-user /opt/nexus /opt/sonatype-work  
sudo chmod -R 755 /opt/nexus /opt/sonatype-work  
[ec2-user@ip-172-31-28-159 ~]$ cd /opt/nexus/bin/  
[ec2-user@ip-172-31-28-159 bin]$ ./nexus start  
Starting nexus  
[ec2-user@ip-172-31-28-159 bin]$ ./nexus status  
nexus is running.  
[ec2-user@ip-172-31-28-159 bin]$ |
```

Copy the nexus ip:8081 to check it in browser.

Jenkins Task

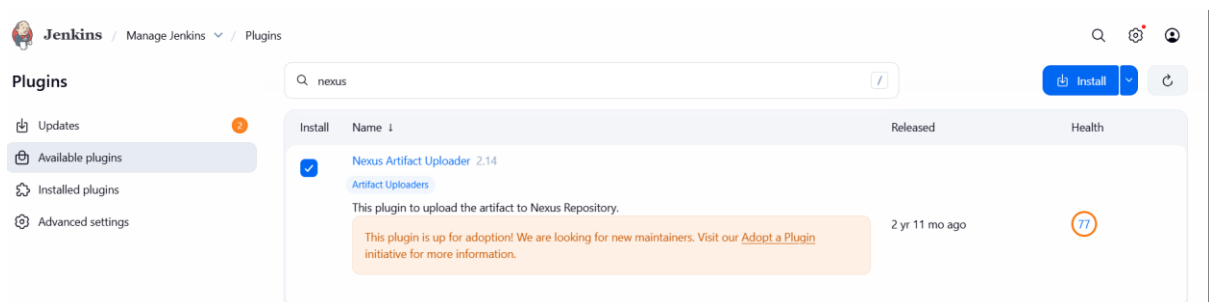


After login the username is admin and password of it you can find as below path and cat and see the password as we done in Jenkins.

My password:-dce0e95f-64bb-4b6a-b587-75047f2cd104

```
ec2-user@ip-172-31-28-159 ~]$ sudo su -
Last login: Thu Oct 30 12:38:22 UTC 2025 on pts/3
[root@ip-172-31-28-159 ~]# /opt/sonatype-work/nexus3/admin.password
-bash: /opt/sonatype-work/nexus3/admin.password: Permission denied
[root@ip-172-31-28-159 ~]# cd /opt/
[root@ip-172-31-28-159 opt]# ls
aws  nexus  nexus-3.68.1-02-unix.tar.gz  sonatype-work
[root@ip-172-31-28-159 opt]# cd sonatype-work/
[root@ip-172-31-28-159 sonatype-work]# ls
nexus3
[root@ip-172-31-28-159 sonatype-work]# cd nexus3/
[root@ip-172-31-28-159 nexus3]# ls
admin.password  cache  elasticsearch  generated-bundles  kar  keystores  log  port  tmp
plobs          db      etc            instances          karaf.pid  lock     orient  restore-from-backup
[root@ip-172-31-28-159 nexus3]# vi admin.password
[root@ip-172-31-28-159 nexus3]#
```

After successful login with credentials .Now, go to Jenkins and download the nexus plugin.



Add nexus artifact uploader in your job/Task.

Jenkins Task

Jenkins

Task-03

Configuration

Search

Settings

User

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

MVN_HOME

Goals

clean install

Advanced

+ Add build step

Filter

Execute SonarQube Scanner

Execute Windows batch command

Execute shell

Invoke Ant

Invoke Gradle script

Invoke top-level Maven targets

Nexus artifact uploader

Run with timeout

Set build status to "pending" on GitHub commit

Some SonarScanner for Maven 3.3.1 - Build Analysis

sending notifications, archiving artifacts, or triggering other jobs.

Nexus artifact uploader

×

Nexus Details

Nexus Version

NEXUS3

Protocol

HTTP

Nexus URL ?

65.0.116.164:8081

Credentials

admin/***** (Nexus-server)

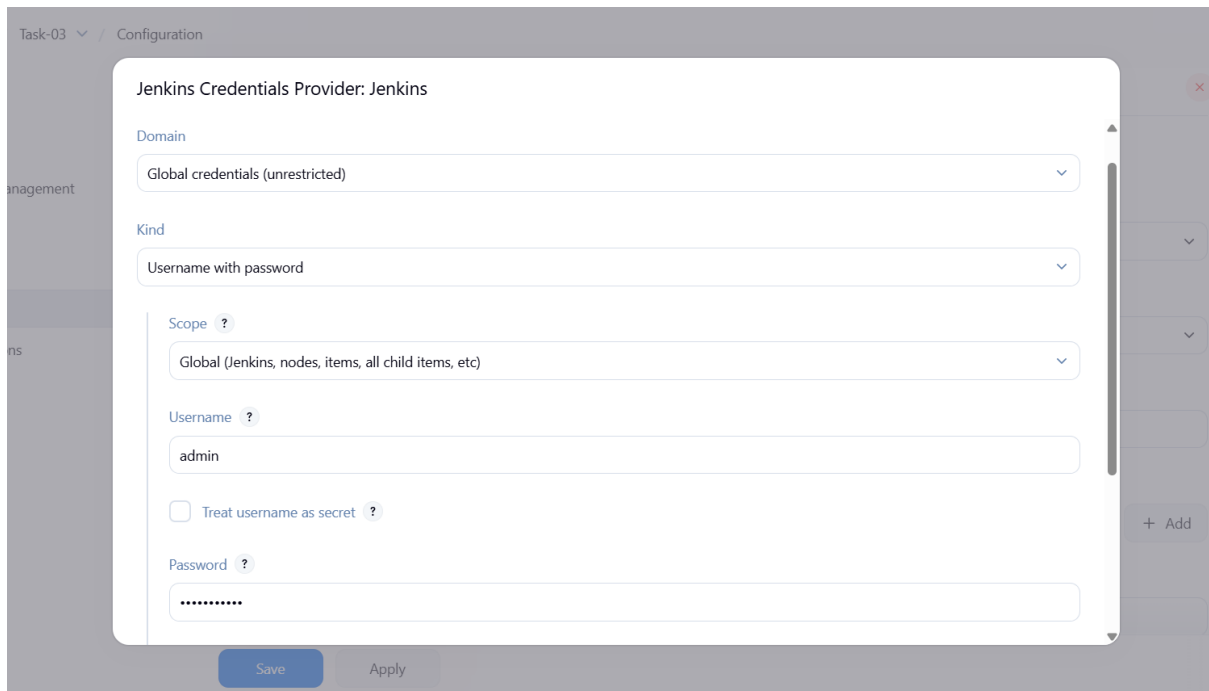
+ Add

GroupId

Save

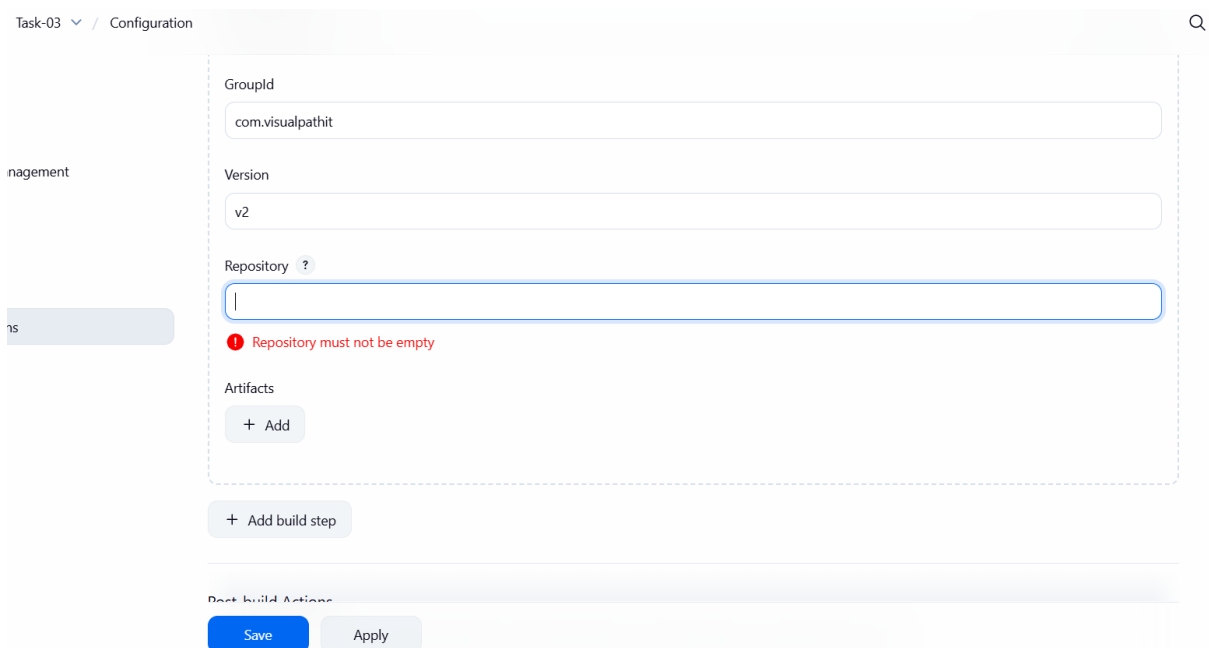
Apply

Jenkins Task



The screenshot shows the 'Jenkins Credentials Provider: Jenkins' configuration window. It includes fields for 'Domain' (Global credentials (unrestricted)), 'Kind' (Username with password), 'Scope' (Global (Jenkins, nodes, items, all child items, etc)), 'Username' (admin), and 'Password' (masked with dots). There is a checkbox for 'Treat username as secret' and 'Save' and 'Apply' buttons at the bottom.

Coming to groupid and version we can find them in our git hub repository in pom.xml file itself. Copy and paste the same here.

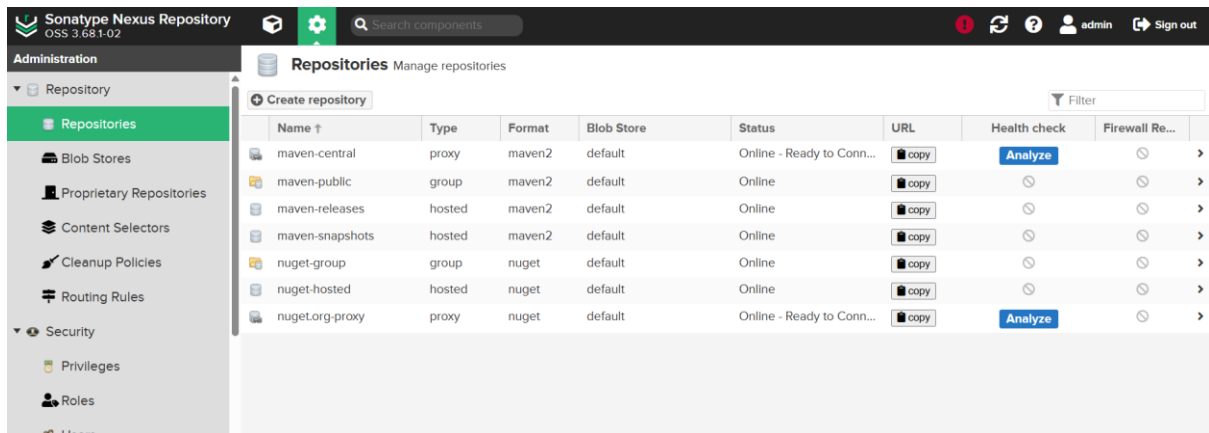


The screenshot shows the 'Task-03 / Configuration' page. It includes fields for 'GroupId' (com.visualpathit), 'Version' (v2), and 'Repository' (empty). A red error message 'Repository must not be empty' is displayed below the 'Repository' field. There is an 'Artifacts' section with a '+ Add' button and a '+ Add build step' button at the bottom. 'Save' and 'Apply' buttons are at the bottom.

Coming to repository we have to create it in nexus itself as we did not create any till now.

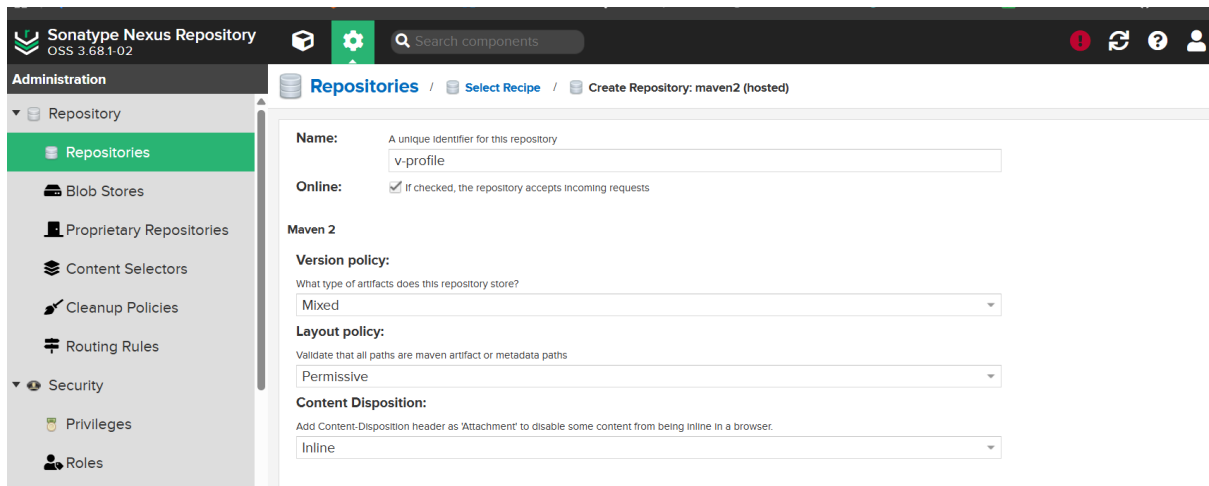
Go to nexus -> click on settings icon on the top -> select repositories->create repository->select maven2 (hosted).

Jenkins Task



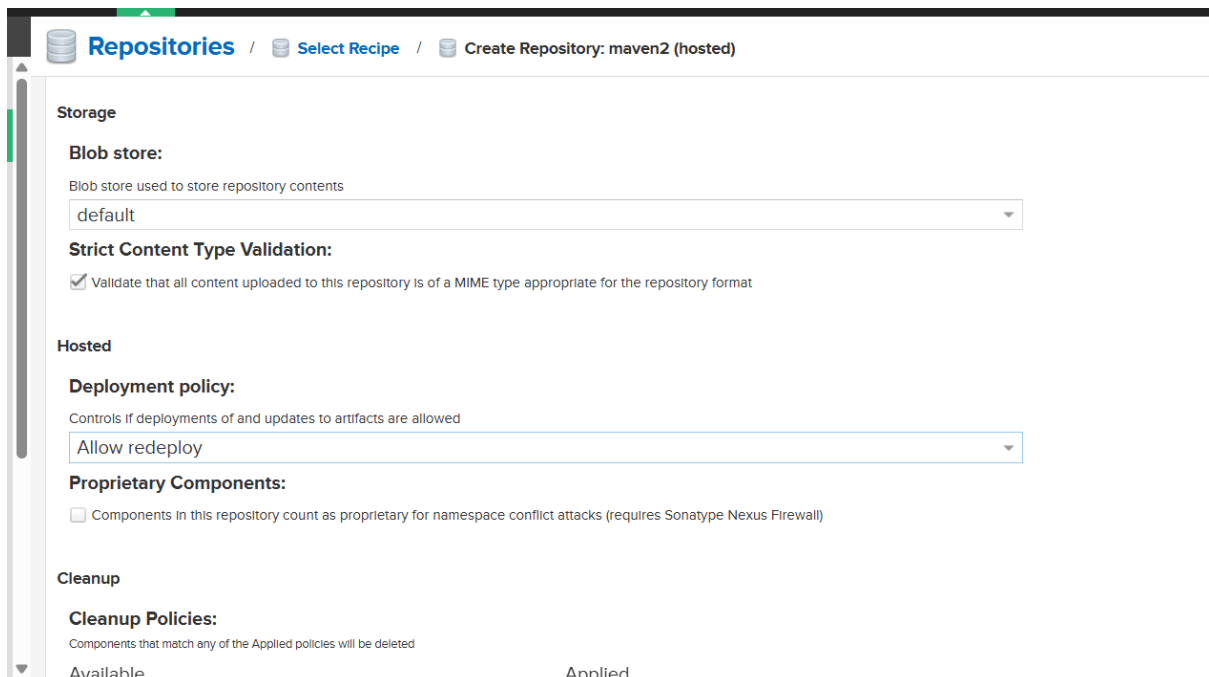
The screenshot shows the Sonatype Nexus Repository OSS 3.68.1-02 interface. The left sidebar contains the 'Administration' menu with 'Repositories' selected. The main area displays a table of existing repositories.

Name	Type	Format	Blob Store	Status	URL	Health check	Firewall Re...
maven-central	proxy	maven2	default	Online - Ready to Conn...		Analyze	
maven-public	group	maven2	default	Online			
maven-releases	hosted	maven2	default	Online			
maven-snapshots	hosted	maven2	default	Online			
nuget-group	group	nuget	default	Online			
nuget-hosted	hosted	nuget	default	Online			
nuget.org-proxy	proxy	nuget	default	Online - Ready to Conn...		Analyze	



The screenshot shows the 'Create Repository: maven2 (hosted)' configuration page. The left sidebar is the same as the previous screenshot. The main area contains a form with the following fields:

- Name:** A unique identifier for this repository. Value: v-profile
- Online:** ☒ If checked, the repository accepts incoming requests
- Maven 2**
 - Version policy:** What type of artifacts does this repository store? Value: Mixed
 - Layout policy:** Validate that all paths are maven artifact or metadata paths. Value: Permissive
 - Content Disposition:** Add Content-Disposition header as 'Attachment' to disable some content from being inline in a browser. Value: Inline

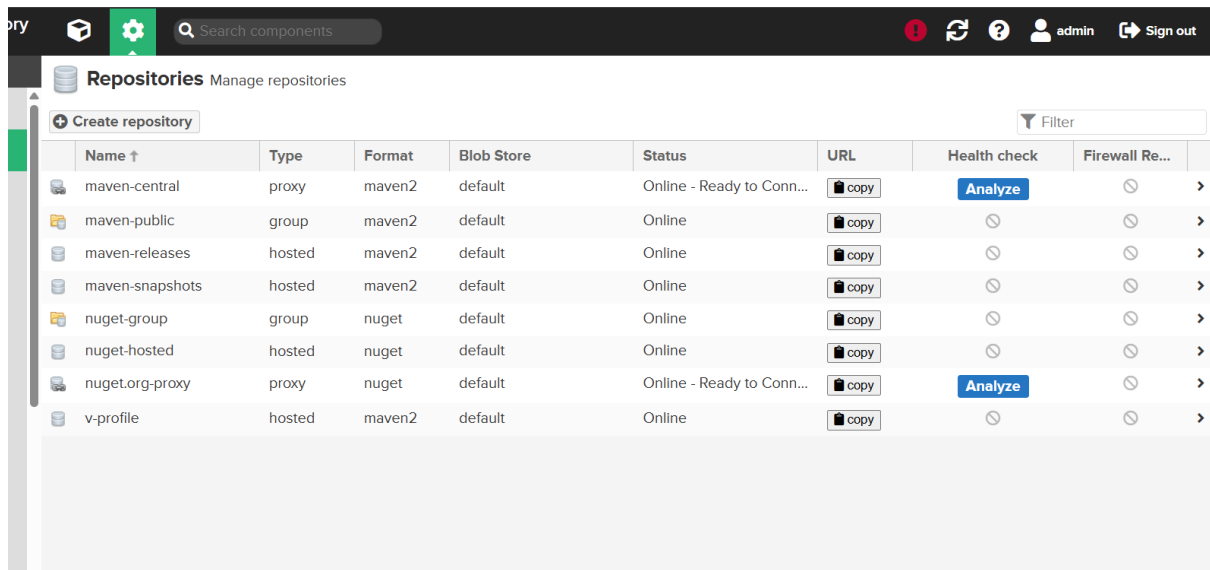


The screenshot shows the 'Create Repository: maven2 (hosted)' configuration page, specifically the 'Storage' and 'Hosted' sections. The left sidebar is the same as the previous screenshots. The main area contains the following settings:

- Storage**
 - Blob store:** Blob store used to store repository contents. Value: default
 - Strict Content Type Validation:** ☒ Validate that all content uploaded to this repository is of a MIME type appropriate for the repository format
- Hosted**
 - Deployment policy:** Controls if deployments of and updates to artifacts are allowed. Value: Allow redeploy
 - Proprietary Components:** ☐ Components in this repository count as proprietary for namespace conflict attacks (requires Sonatype Nexus Firewall)
- Cleanup**
 - Cleanup Policies:** Components that match any of the Applied policies will be deleted

Create.Now we can see our repository in dashboard.

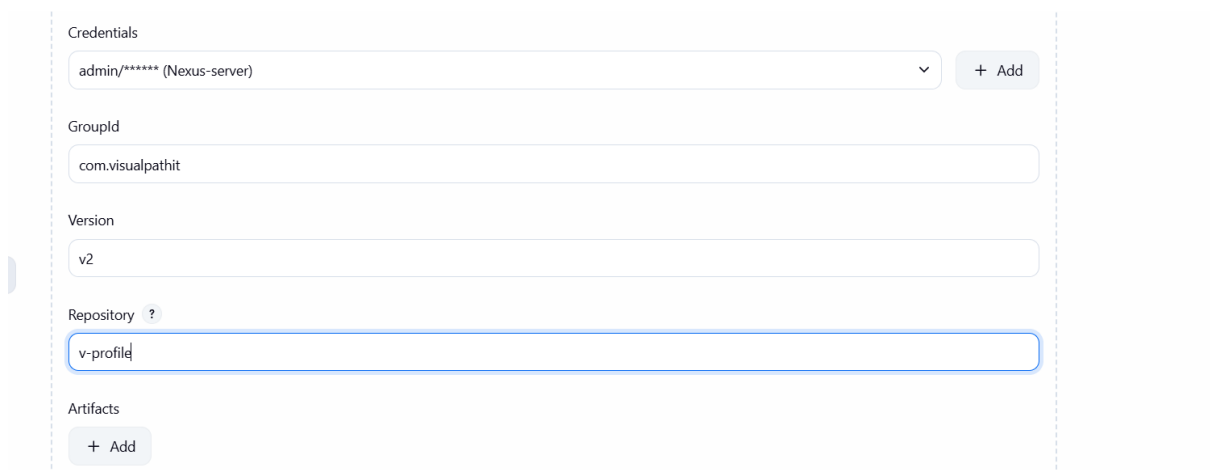
Jenkins Task



The screenshot shows the Jenkins 'Repositories' page. At the top, there's a header with 'Repositories Manage repositories' and a 'Create repository' button. Below this is a table with columns: Name, Type, Format, Blob Store, Status, URL, Health check, and Firewall Re... The table lists several repositories, including 'maven-central', 'maven-public', 'maven-releases', 'maven-snapshots', 'nuget-group', 'nuget-hosted', 'nuget.org-proxy', and 'v-profile'. The 'v-profile' repository is highlighted in blue. It has a status of 'Online' and a health check button labeled 'Analyze'.

Name ↑	Type	Format	Blob Store	Status	URL	Health check	Firewall Re...
maven-central	proxy	maven2	default	Online - Ready to Conn...	copy	Analyze	⌵
maven-public	group	maven2	default	Online	copy	⌵	⌵
maven-releases	hosted	maven2	default	Online	copy	⌵	⌵
maven-snapshots	hosted	maven2	default	Online	copy	⌵	⌵
nuget-group	group	nuget	default	Online	copy	⌵	⌵
nuget-hosted	hosted	nuget	default	Online	copy	⌵	⌵
nuget.org-proxy	proxy	nuget	default	Online - Ready to Conn...	copy	Analyze	⌵
v-profile	hosted	maven2	default	Online	copy	⌵	⌵

Now, copy the name of the repository in my case it is v-profile. And paste it in Jenkins.



The screenshot shows the Jenkins 'Credentials' form. It has several sections: 'Credentials' with a dropdown menu showing 'admin/***** (Nexus-server)' and a '+ Add' button; 'GroupId' with a text field containing 'com.visualpathit'; 'Version' with a text field containing 'v2'; 'Repository' with a text field containing 'v-profile'; and 'Artifacts' with a '+ Add' button.

Credentials

admin/***** (Nexus-server) [+ Add](#)

GroupId

com.visualpathit

Version

v2

Repository ?

v-profile

Artifacts

[+ Add](#)

Now, Add artifacts from your github repository itself in the pom.xml file.

Jenkins Task

Artifacts

Artifact

ArtifactId

vprofile

Type ?

war

Classifier ?

File ?

*.war

Save Apply

After saving and trying to build the job it fails. Issue is the *.war cant be located so see output generated and copy the path where it is getting built.

/var/lib/jenkins/workspace/Task-03/target/vprofile-v2.war

Copy and paste it in file inplace of *.war

```
[INFO] --- war:3.4.0:war (default-war) @ vprofile ---
[INFO] Packaging webapp
[INFO] Assembling webapp [vprofile] in [/var/lib/jenkins/workspace/Task-03/target/vprofile-v2]
[INFO] Processing war project
[INFO] Copying webapp resources [/var/lib/jenkins/workspace/Task-03/src/main/webapp]
[INFO] Building war: /var/lib/jenkins/workspace/Task-03/target/vprofile-v2.war
[INFO] --- install:3.1.2:install (default-install) @ vprofile ---
[INFO] Installing /var/lib/jenkins/workspace/Task-03/pom.xml to /var/lib/jenkins/.m2/repository/com/visualpathit/vprofile/v2/vprofile-v2.pom
[INFO] Installing /var/lib/jenkins/workspace/Task-03/target/vprofile-v2.war to /var/lib/jenkins/.m2/repository/com/visualpathit/vprofile/v2/vprofile-v2.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
```

Task-03 / Configuration

Configuration

vprofile

Type ?

war

Classifier ?

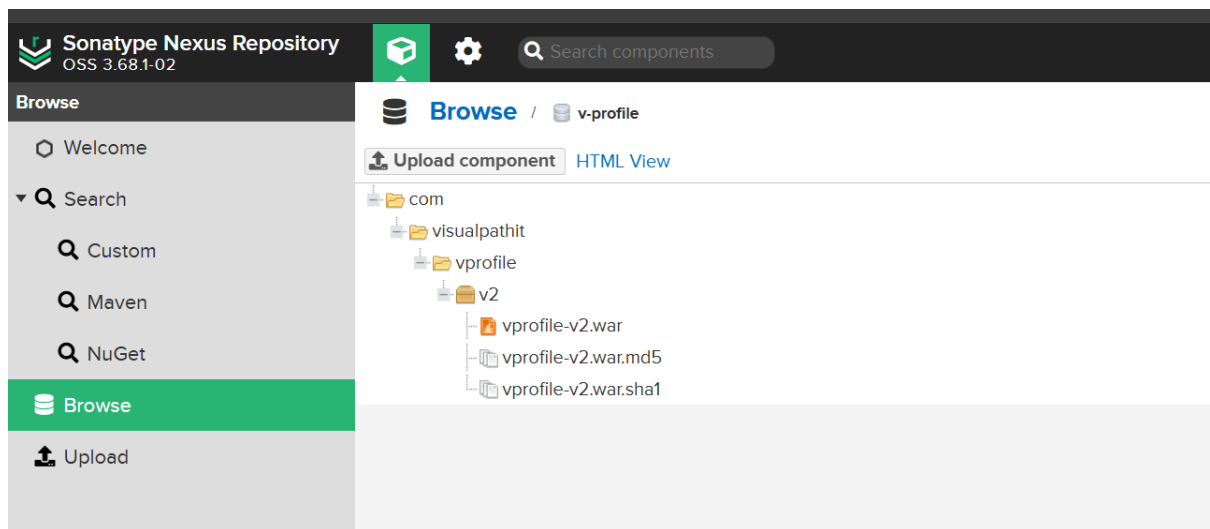
File ?

/var/lib/jenkins/workspace/Task-03/target/vprofile-v2.war

Jenkins Task

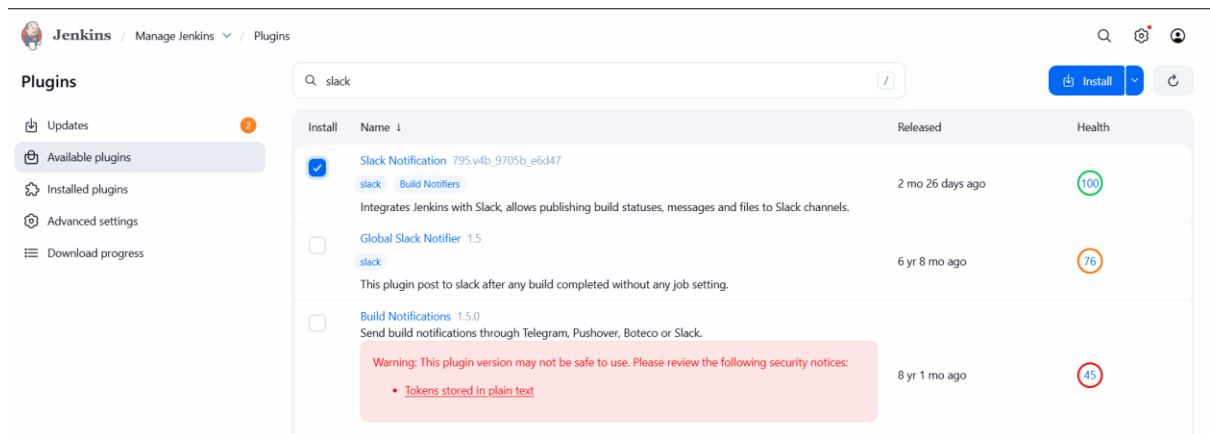
```
ins / Task-03 / #20 / Console Output

UPLOADED: com.visualpathit:vprofile-v2:war:1.0.0
GroupId: com.visualpathit
ArtifactId: vprofile
Classifier:
Type: war
Version: v2
File: vprofile-v2.war
Repository: v-profile
Uploading: http://65.0.116.164:8081/repository/v-profile/com/visualpathit/vprofile/v2/vprofile-v2.war
10 % completed (1.9 MB / 17 MB).
20 % completed (3.6 MB / 17 MB).
30 % completed (5.3 MB / 17 MB).
40 % completed (7.1 MB / 17 MB).
50 % completed (8.8 MB / 17 MB).
60 % completed (10 MB / 17 MB).
70 % completed (12 MB / 17 MB).
80 % completed (14 MB / 17 MB).
90 % completed (16 MB / 17 MB).
100 % completed (17 MB / 17 MB).
Uploaded: http://65.0.116.164:8081/repository/v-profile/com/visualpathit/vprofile/v2/vprofile-v2.war (17 MB at 26 MB/s)
Uploading artifact vprofile-v2.war completed.
Finished: SUCCESS
```

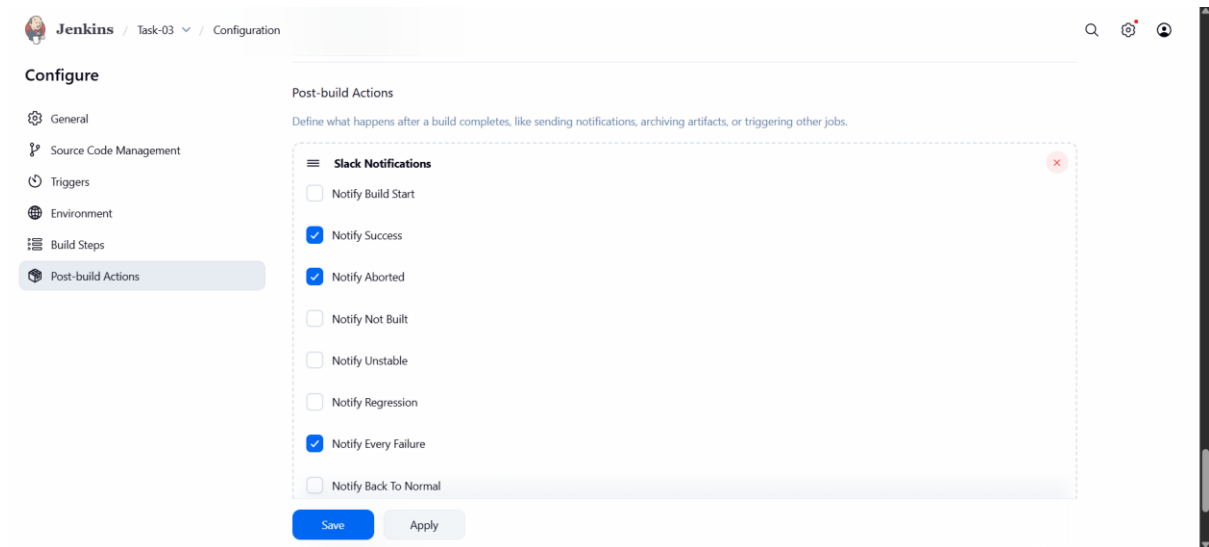


We can get multiple versions here itself. If we run our built it will be generated as new version .We can have here multiple versions too.

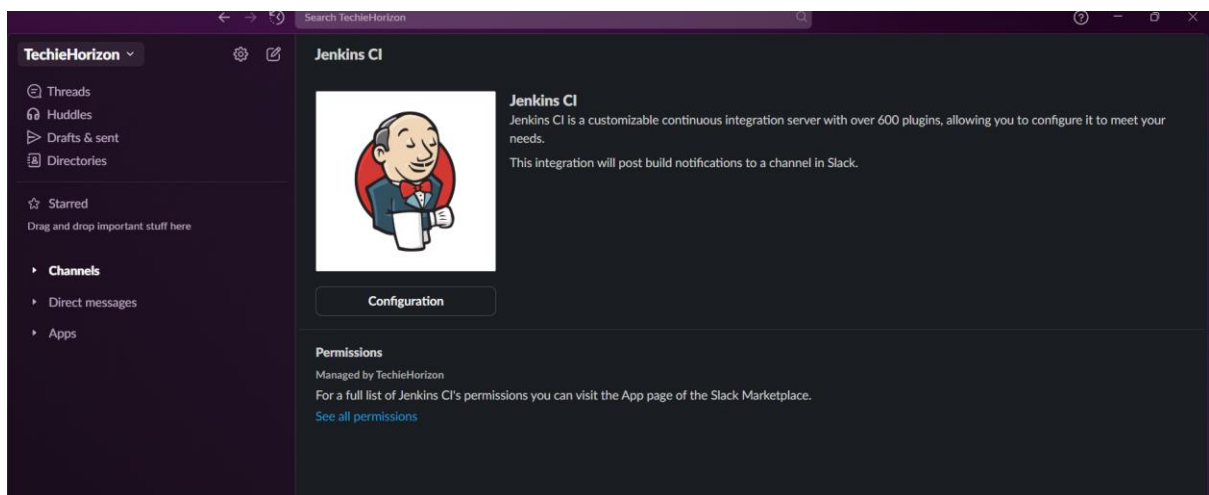
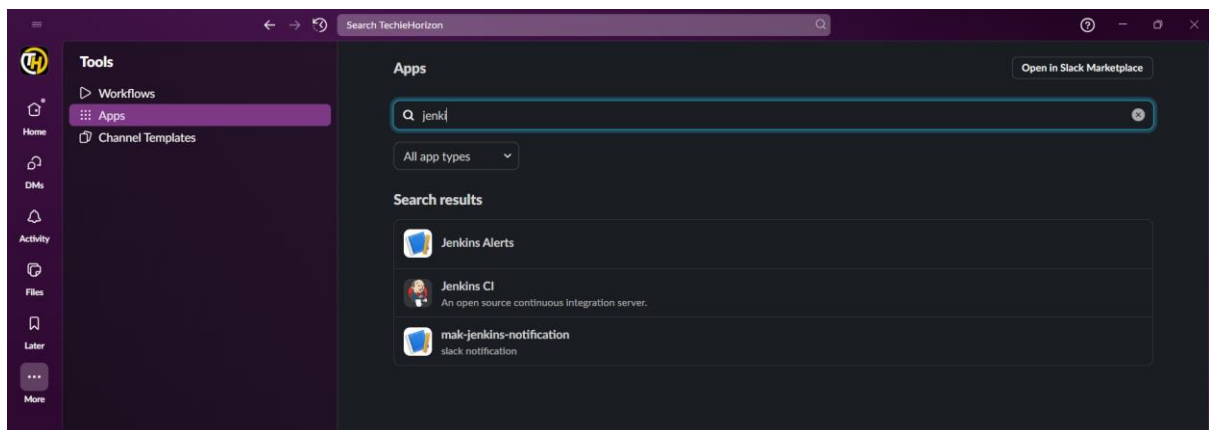
SLACK INTEGRATION: -



Jenkins Task



Now go to slack.



Jenkins Task

 **slack**
from Salesforce

BrowseCategories ▾Collections ▾ManageBuild

Find an app or a service you already use

TechieHorizon ▾

[Browse apps](#) > [Jenkins CI](#) > [New configuration](#)**Jenkins CI**
An open source continuous integration server.Jenkins CI is a customizable continuous integration server with over 600 plugins, allowing you to configure it to meet your needs.
This integration will post build notifications to a channel in Slack.

Post to Channel
Start by choosing a channel where
Jenkins notifications will be posted.

#jenkins-integration ▾

Add Jenkins CI Integration

After adding channel you will get a document how to setup with Jenkins go through the document.

 **Jenkins** / [Manage Jenkins](#) ▾ / [System](#) ▾

Search ⌕ Settings ⚙️ Profile 👤

Slack

Workspace ?
techiehorizon

Credential ?
slack-ID ▾ + Add

Default channel / member id ?
jenkins-integration

☐ Custom slack app bot user ?

Advanced ^ [Edited](#)

Icon Emoji ?

Save Apply

Jenkins Task

Jenkins Credentials Provider: Jenkins

Global credentials (unrestricted)

Kind

Secret text

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

Secret

.....

ID ?

slack-ID

Description ?

slack-ID

Apply

+ Add

Test Connection

Clear cache

Jenkins / Manage Jenkins / System

Advanced Edited

Icon Emoji ?

Username ?

salvarun-15

☐ Send as text ?

Override url ?

User ID Resolver

☐ No User ID Resolver ?

☐ Slack email User ID Resolver ?

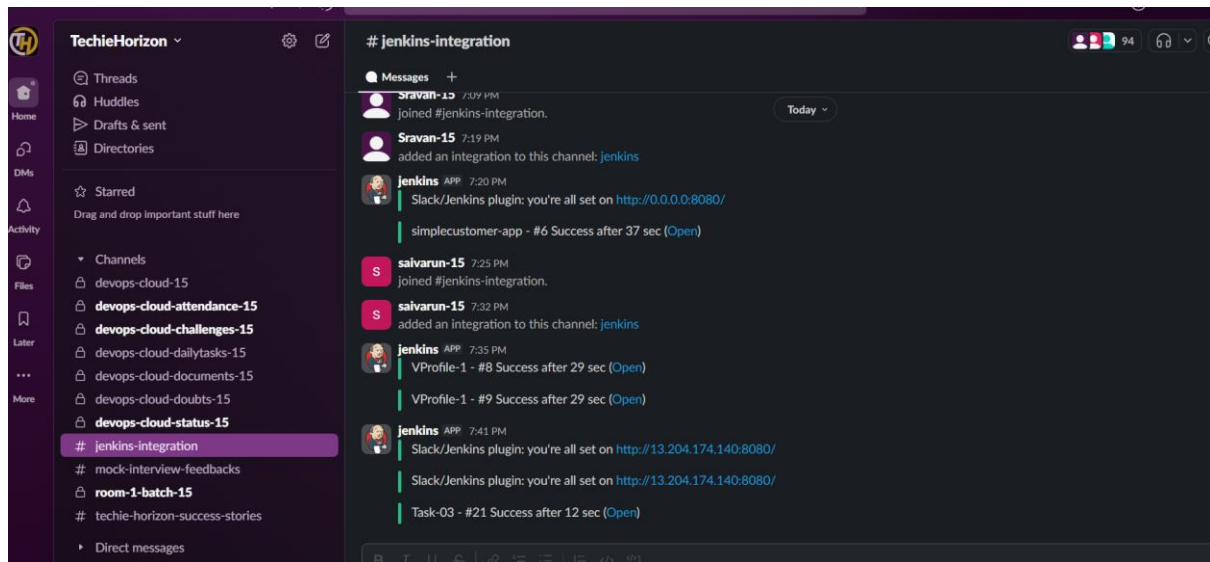
Success

Test Connection

Save Apply

After saving and built the task you will receive a notification in the channel.

Jenkins Task

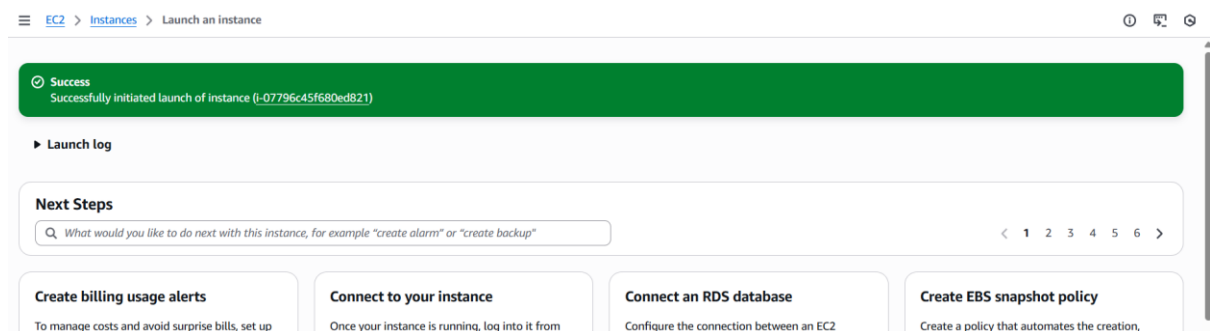


TOMCAT DEPLOYMENT:-

Add deploy to container plugin in your Jenkins.



Launch one more instance for tomcat.



Install java in it.

Jenkins Task

[illegible]

Now download tomcat in your instance.

<https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.111/bin/apache-tomcat-9.0.111.tar.gz>

```
xml-common-0.6.3-56.amzn2023.0.2.noarch
xprop-1.2.7-1.amzn2023.x86_64

Complete!
[root@ip-172-31-25-4 ~]# java -version
openjdk version "1.8.0_472"
OpenJDK Runtime Environment corretto-8.472.08.1 (build 1.8.0_472-b08)
OpenJDK 64-Bit Server VM Corretto-8.472.08.1 (build 25.472-b08, mixed mode)
[root@ip-172-31-25-4 ~]# cd /opt/
[root@ip-172-31-25-4 opt]# wget https://d1cdn.apache.org/tomcat/tomcat-9/v9.0.111/bin/apache-tomcat-9.0.111.tar.gz
```

Now extract

```
[root@ip-172-31-25-4 opt]# ls
apache-tomcat-9.0.111.tar.gz  aws
[root@ip-172-31-25-4 opt]# tar xvf a
apache-tomcat-9.0.111.tar.gz  aws/
[root@ip-172-31-25-4 opt]# tar xvf apache-tomcat-9.0.111.tar.gz
```

Now start tomcat

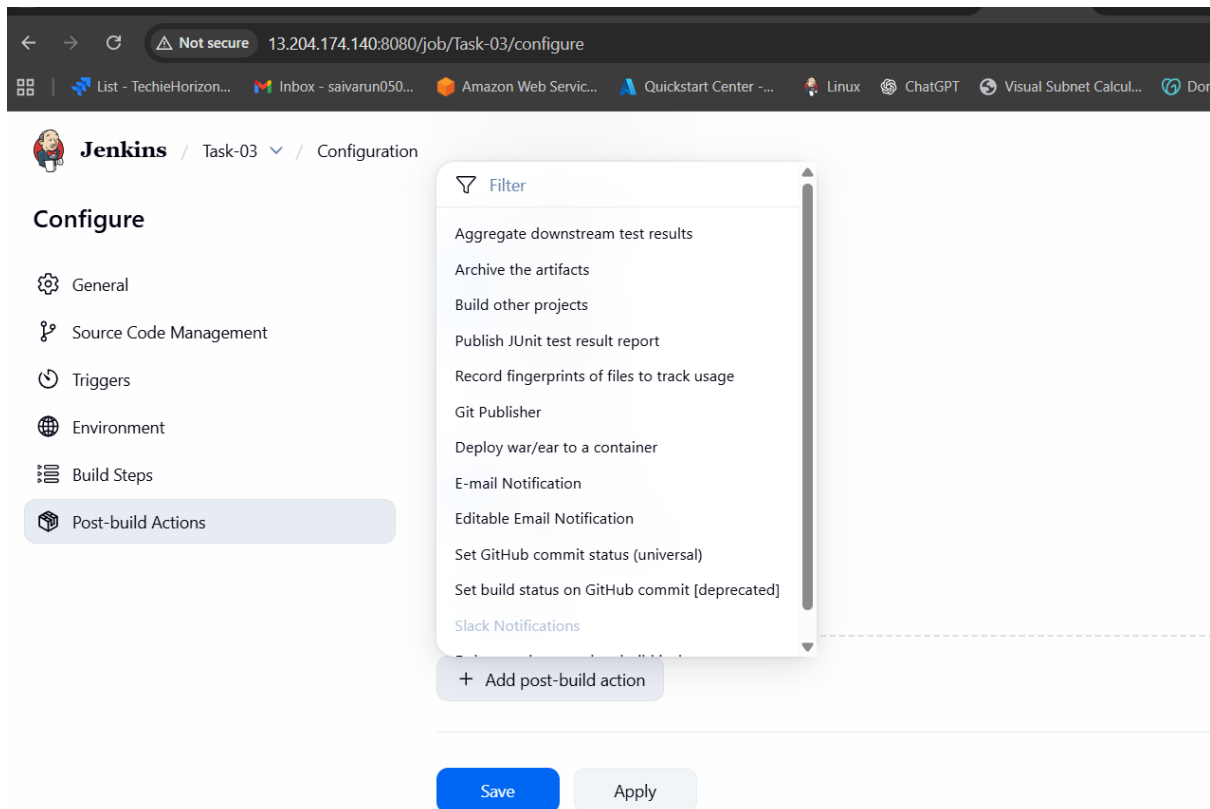
```

apache-tomcat-9.0.111/bin/cool-wippler.sh
apache-tomcat-9.0.111/bin/version.sh
[root@ip-172-31-25-4 opt]# cd apache-tomcat-9.0.111/
[root@ip-172-31-25-4 apache-tomcat-9.0.111]# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@ip-172-31-25-4 apache-tomcat-9.0.111]# cd bin/
[root@ip-172-31-25-4 bin]# bash startup.sh
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.111
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.111
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.111/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.111/bin/bootstrap.jar:/opt/apache-tomcat-9.0.111/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-31-25-4 bin]# |

```

Jenkins Task

Now go to Jenkins job and configure. Go to post build and select deploy war/ear to a container.



Now go to the below path and comment the files.

```
USING CATALINA_OPTS:
Tomcat started.
[root@ip-172-31-25-4 bin]# find / -name context.xml
/opt/apache-tomcat-9.0.111/conf/context.xml
/opt/apache-tomcat-9.0.111/webapps/docs/META-INF/context.xml
/opt/apache-tomcat-9.0.111/webapps/examples/META-INF/context.xml
/opt/apache-tomcat-9.0.111/webapps/host-manager/META-INF/context.xml
/opt/apache-tomcat-9.0.111/webapps/manager/META-INF/context.xml
[root@ip-172-31-25-4 bin]# |
```

Open each file and add comment as shown <!-- -->

Jenkins Task

```
root@ip-172-31-25-4:/opt/apache-tomcat-9.0.111/bin
<?xml version="1.0" encoding="UTF-8"?>
<!--
Licensed to the Apache Software Foundation (ASF) under one or more
contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License. You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<Context antiResourceLocking="false" >
    <!-- <Valve className="org.apache.catalina.valves.RemoteCIDRValve"
        allow="127.0.0.0/8,:::1/128" /> -->
</Context>
~
~
~
~
~
~
~
~
```

Do same for all the 4 files.

```
tomcat started.
[root@ip-172-31-25-4 bin]# find / -name context.xml
/opt/apache-tomcat-9.0.111/conf/context.xml
/opt/apache-tomcat-9.0.111/webapps/docs/META-INF/context.xml
/opt/apache-tomcat-9.0.111/webapps/examples/META-INF/context.xml
/opt/apache-tomcat-9.0.111/webapps/host-manager/META-INF/context.xml
/opt/apache-tomcat-9.0.111/webapps/manager/META-INF/context.xml
[root@ip-172-31-25-4 bin]# vi /opt/apache-tomcat-9.0.111/webapps/docs/META-INF/context.xml
[root@ip-172-31-25-4 bin]# vi /opt/apache-tomcat-9.0.111/webapps/examples/META-INF/context.xml
[root@ip-172-31-25-4 bin]# vi /opt/apache-tomcat-9.0.111/webapps/host-manager/META-INF/context.xml
[root@ip-172-31-25-4 bin]# vi /opt/apache-tomcat-9.0.111/webapps/manager/META-INF/context.xml
[root@ip-172-31-25-4 bin]# |
```

Update users information in the tomcat-users.xml file

goto tomcat home directory and Add below users to conf/tomcat-user.xml file

```
<role rolename="manager-gui"/>

<role rolename="manager-script"/>

<role rolename="manager-jmx"/>

<role rolename="manager-status"/>

<user username="admin" password="admin" roles="manager-gui, manager-script, manager-
jmx, manager-status"/>

<user username="deployer" password="deployer" roles="manager-script"/>

<user username="tomcat" password="s3cret" roles="manager-gui"/>
```

Jenkins Task

```
[root@ip-172-31-25-4 conf]# ebapps/manager/META-INF/context.xml
[root@ip-172-31-25-4 bin]# vi /opt/apache-tomcat-9.0.111/webapps/docs/META-INF/context.xml
[root@ip-172-31-25-4 bin]# vi /opt/apache-tomcat-9.0.111/webapps/examples/META-INF/context.xml
[root@ip-172-31-25-4 bin]# vi /opt/apache-tomcat-9.0.111/webapps/host-manager/META-INF/context.xml
[root@ip-172-31-25-4 bin]# vi /opt/apache-tomcat-9.0.111/webapps/manager/META-INF/context.xml
[root@ip-172-31-25-4 bin]# cd ..
[root@ip-172-31-25-4 apache-tomcat-9.0.111]# ls
BUILDING.txt  LICENSE  README.md  RUNNING.txt  conf  logs  webapps
CONTRIBUTING.md  NOTICE  RELEASE-NOTES  bin  lib  temp  work
[root@ip-172-31-25-4 apache-tomcat-9.0.111]# cd conf/
[root@ip-172-31-25-4 conf]# ls
Catalina      context.xml      logging.properties  tomcat-users.xsd
catalina.policy  jaspic-providers.xml  server.xml          web.xml
catalina.properties  jaspic-providers.xsd  tomcat-users.xml
[root@ip-172-31-25-4 conf]# vi tomcat-users.xml
```

```
<user username="tomcat" password="<must-be-changed>" roles="tomcat" />
<user username="both" password="<must-be-changed>" roles="tomcat,role1" />
<user username="role1" password="<must-be-changed>" roles="role1" />
-->
<role rolename="manager-gui" />
<role rolename="manager-script" />
<role rolename="manager-jmx" />
<role rolename="manager-status" />
<user username="admin" password="admin" roles="manager-gui, manager-script, manager-jmx, manager-status" />
<user username="deployer" password="deployer" roles="manager-script" />
<user username="tomcat" password="s3cret" roles="manager-gui" />
</tomcat-users>
-- INSERT (paste) --
```

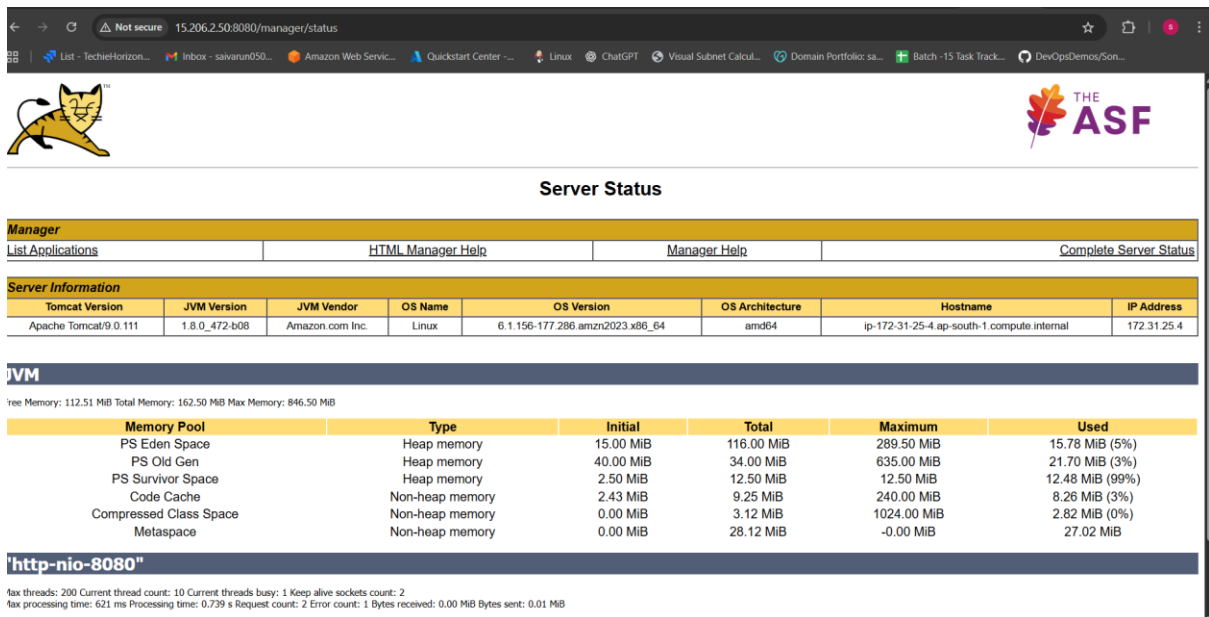
62, 6

Now save and restart tomcat

```
root@ip-172-31-25-4 /opt/apache-tomcat-9.0.111/bin
[root@ip-172-31-25-4 ~]# cd /conf/
-bash: cd: /conf/: No such file or directory
[root@ip-172-31-25-4 ~]# cd /opt/
[root@ip-172-31-25-4 opt]# ls
apache-tomcat-9.0.111  apache-tomcat-9.0.111.tar.gz  aws
[root@ip-172-31-25-4 opt]# cd apache-tomcat-9.0.111/
[root@ip-172-31-25-4 apache-tomcat-9.0.111]# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@ip-172-31-25-4 apache-tomcat-9.0.111]# cd bin/
[root@ip-172-31-25-4 bin]# bash shutdown.sh
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.111
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.111
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.111/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/apache-tomcat-9.0.111/bin/bootstrap.jar:/opt/apache-tomcat-9.0.111/bin/tomcat-juli.jar
[root@ip-172-31-25-4 bin]# bash startup.sh
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.111
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.111
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.111/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/apache-tomcat-9.0.111/bin/bootstrap.jar:/opt/apache-tomcat-9.0.111/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-31-25-4 bin]#
```

Now, refresh the tomcat on browser. And check server status it will ask for username and password
Keep it as deployer, deployer both.

Jenkins Task



The screenshot shows the ASF Manager web interface. At the top, there's a navigation bar with a cat logo on the left and the ASF logo on the right. Below the navigation bar is a "Server Status" section. This section includes a "Manager" tab with links for "List Applications", "HTML Manager Help", "Manager Help", and "Complete Server Status". Below this is a "Server Information" table with columns for Tomcat Version, JVM Version, JVM Vendor, OS Name, OS Version, OS Architecture, Hostname, and IP Address. The table shows details for Apache Tomcat/9.0.111 running on Linux. Below the server information is a "JVM" section with a table showing memory pool details. The table has columns for Memory Pool, Type, Initial, Total, Maximum, and Used. It lists various memory pools like PS Eden Space, PS Old Gen, PS Survivor Space, Code Cache, Compressed Class Space, and Metaspace. At the bottom, there's a section for "http-nio-8080" showing thread and request statistics.

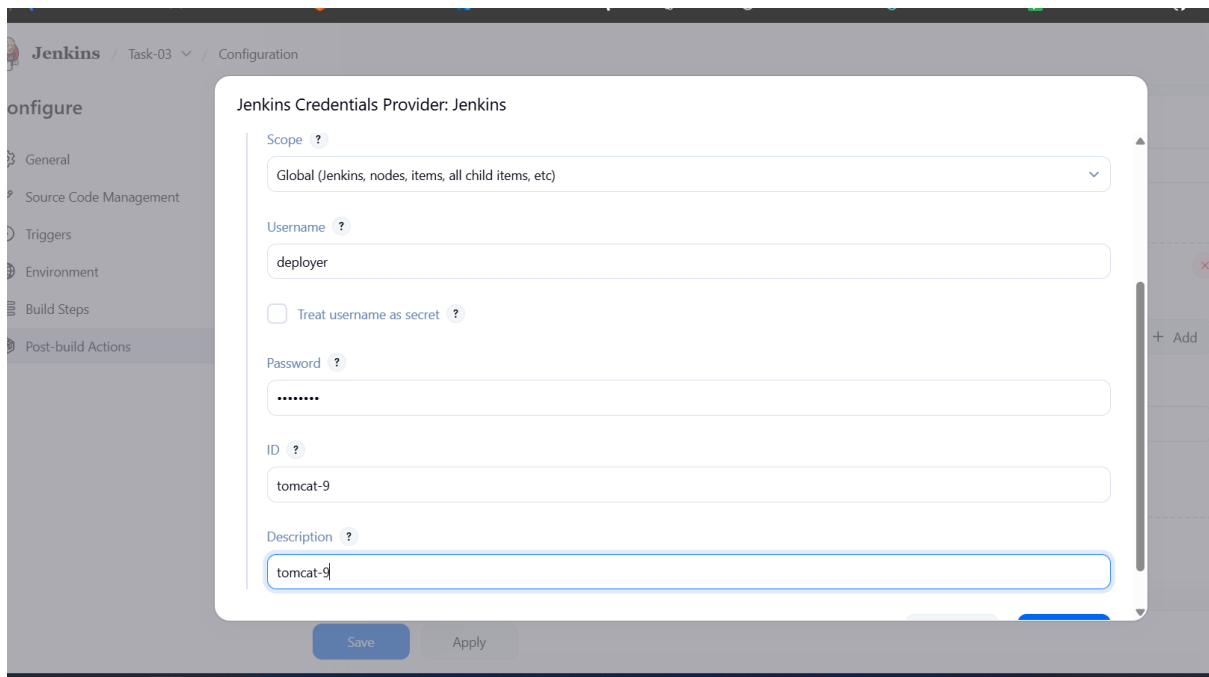
Tomcat Version	JVM Version	JVM Vendor	OS Name	OS Version	OS Architecture	Hostname	IP Address
Apache Tomcat/9.0.111	1.8.0_472-b08	Amazon.com Inc.	Linux	6.1.156-177.286.amzn2023.x86_64	amd64	ip-172-31-25-4.ap-south-1.compute.internal	172.31.25.4

Memory Pool	Type	Initial	Total	Maximum	Used
PS Eden Space	Heap memory	15.00 MiB	116.00 MiB	289.50 MiB	15.78 MiB (5%)
PS Old Gen	Heap memory	40.00 MiB	34.00 MiB	635.00 MiB	21.70 MiB (3%)
PS Survivor Space	Heap memory	2.50 MiB	12.50 MiB	12.50 MiB	12.48 MiB (99%)
Code Cache	Non-heap memory	2.43 MiB	9.25 MiB	240.00 MiB	8.26 MiB (3%)
Compressed Class Space	Non-heap memory	0.00 MiB	3.12 MiB	1024.00 MiB	2.82 MiB (0%)
Metaspace	Non-heap memory	0.00 MiB	28.12 MiB	-0.00 MiB	27.02 MiB

http-nio-8080

Max threads: 200 Current thread count: 10 Current threads busy: 1 Keep alive sockets count: 2
Max processing time: 621 ms Processing time: 0.739 s Request count: 2 Error count: 1 Bytes received: 0.00 MiB Bytes sent: 0.01 MiB

Now add the same key in Jenkins job too.



The screenshot shows the Jenkins Configuration page for a Credentials Provider. The page is titled "Jenkins Credentials Provider: Jenkins". It has a "Scope" dropdown menu set to "Global (Jenkins, nodes, items, all child items, etc)". Below this is a "Username" field with the value "deployer". There is a checkbox for "Treat username as secret" which is unchecked. Below that is a "Password" field with a masked value "*****". Below the password field is an "ID" field with the value "tomcat-9". Below the ID field is a "Description" field with the value "tomcat-9". At the bottom of the configuration form are "Save" and "Apply" buttons.

Jenkins Credentials Provider: Jenkins

Scope ?
Global (Jenkins, nodes, items, all child items, etc)

Username ?
deployer

☐ Treat username as secret ?

Password ?

ID ?
tomcat-9

Description ?
tomcat-9

Save Apply

Jenkins Task



Server Status

[Manager](#)[list Applications](#)[HTML Manager Help](#)[Manager Help](#)[Complete Server Status](#)

Server Information

Tomcat Version	JVM Version	JVM Vendor	OS Name	OS Version	OS Architecture	Hostname	IP Address
Apache Tomcat/9.0.111	1.8.0_472-b08	Amazon.com Inc.	Linux	6.1.156-177.286.amzn2023.x86_64	amd64	ip-172-31-25-4.ap-south-1.compute.internal	172.31.25.4

VM

Free Memory: 112.51 MiB Total Memory: 162.50 MiB Max Memory: 846.50 MiB

Memory Pool	Type	Initial	Total	Maximum	Used
PS Eden Space	Heap memory	15.00 MiB	116.00 MiB	289.50 MiB	15.78 MiB (5%)
PS Old Gen	Heap memory	40.00 MiB	34.00 MiB	635.00 MiB	21.70 MiB (3%)
PS Survivor Space	Heap memory	2.50 MiB	12.50 MiB	12.50 MiB	12.48 MiB (99%)
Code Cache	Non-heap memory	2.43 MiB	9.25 MiB	240.00 MiB	8.26 MiB (3%)
Compressed Class Space	Non-heap memory	0.00 MiB	3.12 MiB	1024.00 MiB	2.82 MiB (0%)
Metaspace	Non-heap memory	0.00 MiB	28.12 MiB	-0.00 MiB	27.02 MiB

http-nio-8080

1xx threads: 200 Current thread count: 10 Current threads busy: 1 Keep alive sockets count: 2
1xx processing time: 621 ms Processing time: 0.739 s Request count: 2 Error count: 1 Bytes received: 0.00 MiB Bytes sent: 0.01 MiB

```
sing CLASSPATH: /opt/apache-tomcat-9.0.111/bin/bootstrap.jar:
sing CATALINA_OPTS:
omcat started.
root@ip-172-31-25-4 bin]# cd ..
root@ip-172-31-25-4 apache-tomcat-9.0.111]# cd webapps/
root@ip-172-31-25-4 webapps]# ll
total 17108
-rwxr-x---. 3 root root 16384 Oct 30 14:30 ROOT
-rwxr-x---. 16 root root 16384 Oct 30 14:30 docs
-rwxr-x---. 7 root root 99 Oct 30 14:30 examples
-rwxr-x---. 6 root root 79 Oct 30 14:30 host-manager
-rwxr-x---. 6 root root 114 Oct 30 14:30 manager
-rwxr-x---. 5 root root 54 Oct 30 15:03 vprofile-v2
-rw-r-----. 1 root root 17484723 Oct 30 15:03 vprofile-v2.war
root@ip-172-31-25-4 webapps]#
```

devops-cloud-documents-15

devops-cloud-doubts-15

devops-cloud-status-15

jenkins-integration

mock-interview-feedbacks

room-1-batch-15

techie-horizon-success-stories

Direct messages

Task-03 - #21 Success after 12 sec (Open)

jenkins APP 8:30 PM
simplecustomer-app - #7 Success after 38 sec (Open)
simplecustomer-app - #8 Success after 37 sec (Open)

jenkins APP 8:33 PM
Task-03 - #22 Success after 14 sec (Open)